Bridge Completion and an Equivalent Branch-Vertex Formulation of the Induced Cycles Conjecture

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Abstract

For a subcubic graph H, let $b_{\mathrm{ind}}(H)$ be the maximum number of its degree-three vertices contained in an induced 2-regular subgraph. We construct a simple cubic completion $\mathcal{C}(H)$ by attaching a fixed thirteen-vertex graph at every missing incidence. If $D = 3|V(H)| - 2|E(H)|$, then

$$|V(\mathcal{C}(H))| = |V(H)| + 13D, \qquad c_{\mathrm{ind}}(\mathcal{C}(H)) = 7D + b_{\mathrm{ind}}(H).$$

Consequently, the cubic induced cycles conjecture is equivalent to $b_{\mathrm{ind}}(H) \geq |E(H)| - |V(H)|$ for every subcubic graph H. The reduction preserves the exact surplus over the conjectured bound. Further exact formulas describe edge replacements and infinite families attaining the half bound. A matching-polytope argument supplies a sufficient condition for nonuniform replacements over bridgeless cubic bases. Any counterexample containing a bridge would yield a counterexample obtained by joining two copies of one rooted graph. A fractional cut criterion certifies arbitrary triangle-chain replacements over 2,254 bases with checked rational witnesses. A conditional version also certifies the sharp twenty-six-vertex double of the attachment. A smallest counterexample to the branch formulation would be obtained from a simple cubic base by expanding each chosen edge once, with fewer than half as many chosen edges as base vertices. Those chosen edges must form a forest and satisfy an additional sparsity condition on every subset.

We also reformulate good sets as odd spanning factors without isolated edges. An odd pseudoforest of this kind would establish the conjectured bound. We prove a first repair step from every perfect matching, and a complete factor census through order twenty verifies repairs using at most two cycle exchanges. Nevertheless, the proposed monotone repair assertion is false. For every $k \geq 3$, we construct a cubic graph of order $28k - 26$ and an odd pseudoforest with one isolated edge such that every repair must pass through at least k isolated edges. This barrier is attained, and the shortest repair has k exchanges. Each graph in this family has induced cycles vertex number exactly half its order. Thus the obstructions refute the repair strategy while satisfying the original bound. The general conjecture remains unresolved.

1 Problem and scope

All graphs are finite and simple. A graph is *subcubic* if its maximum degree is at most three. A set S is *good* when the induced graph on S is 2-regular; the empty set is allowed. Write $c_{\mathrm{ind}}(G) = \max\{|S| : S \text{ is good in } G\}$. The conjecture of Henning, Joos, Löwenstein and Sasse is

$$c_{\mathrm{ind}}(G) \geq |V(G)|/2 \quad \text{for every cubic graph } G. \tag{1}$$

The original source is [5]; recent papers [6, 7] still leave the general assertion unresolved.

A previous note in this repository proves (1) for graphs with a square factor. Here we study attachments across bridges. The structural results below have direct proofs; finite computational

results are identified separately. Targeted literature searches did not establish publication priority; no claim of priority is made. In particular, introducing b_{ind} is a convenient reformulation of the current problem, not a claimed resolution under new terminology.

2 An attachment with an exact boundary cost

Construct a graph F from three disjoint copies Q_0, Q_1, Q_2 of $K_{2,2}$. In Q_i label the two sides $\{\ell_i^0, \ell_i^1\}$ and $\{r_i^0, r_i^1\}$, and add

$$r_i^j \ell_{i+1}^j \qquad (i \in \mathbb{Z}/3\mathbb{Z},\ j \in \{0,1\}).$$

Thus F has twelve vertices and is cubic. Set $e = r_0^0 \ell_1^0$. Let P be obtained by subdividing e once, with new vertex z. The *root* z has degree two, and all other vertices of P have degree three.

Lemma 2.1. *The largest good sets in P containing and avoiding z have orders seven and six, respectively. Moreover, $V(P)$ can be partitioned into a good seven-set containing z and a good six-set avoiding z.*

Proof. We first show that both F and $F - e$ have induced cycles vertex number six. In either graph, a good set meets a square in zero, two, three, or four vertices. A singleton or an opposite pair cannot occur: each such selected vertex has no selected neighbour in the square and at most one neighbour outside it. A two-set must be adjacent.

If an entire square is selected, its matching neighbours in the other two squares must be unselected. For F this immediately empties both other squares, since their remaining vertices form independent pairs. The same conclusion holds for $F - e$ if the selected square is Q_2. If it is Q_0 or Q_1, one neighbouring square is still forced empty. The last square has one forced zero on its side facing the full square; any possible adjacent pair or triple would need a selected matching neighbour in the empty square. Hence it too is empty. Thus a four-set contributes only four vertices in total.

Suppose no square is full, but some square contains three selected vertices. Its two selected vertices on the same side have internal degree one. Both therefore require selected matching neighbours. Neither required matching edge can be the deleted edge. The neighbouring square contains both vertices on one side; it must therefore also be a three-set, with that side doubled. These two three-sets face each other. On their outward sides, the only selected vertex has internal degree two, so it cannot be adjacent to any further selected vertex. The third square cannot contain an adjacent two-set, and cannot form another facing pair of three-sets. It is empty. The total is six. If no square contains three or four selected vertices, the total is at most $2 + 2 + 2 = 6$.

For each $j = 0, 1$, the six vertices $\{\ell_i^j, r_i^j : i = 0, 1, 2\}$ form an induced cycle in F. The cycle with $j = 1$ also survives in $F - e$. This proves the asserted maxima of six.

A good set in P avoiding z is exactly a good set in $F - e$, so its maximum is six. If it contains z, it contains both endpoints of e. Suppressing z turns it into a good set in F smaller by one, giving an upper bound of seven. Subdivide the $j = 0$ six-cycle to attain seven. The $j = 1$ six-cycle is its vertex-set complement in P, proving the last assertion. □

Every bridge of a graph is absent from every cycle. Consequently, no bridge can occur as an edge of an induced 2-regular subgraph. This elementary fact is essential in the following construction.

3 Exact cubic completion

For a subcubic graph H, put

$$V_3(H) = \{v : d_H(v) = 3\}, \qquad b_{\mathrm{ind}}(H) = \max_{S \text{ good in } H} |S \cap V_3(H)|.$$

Degrees in this definition are measured in H, not in the selected subgraph. Let $n = |V(H)|$, $m = |E(H)|$, and

$$d(v) = 3 - d_H(v), \qquad D = \sum_v d(v) = 3n - 2m.$$

At each vertex v, attach $d(v)$ disjoint copies of P by joining v to their roots. Denote the resulting graph by $\mathcal{C}(H)$. These new edges are bridges. The graph $\mathcal{C}(H)$ is simple and cubic, is connected whenever H is connected, and has order $n + 13D$.

Theorem 3.1 (Exact completion formula)**.** *For every subcubic graph H,*

$$c_{\mathrm{ind}}(\mathcal{C}(H)) = 7D + b_{\mathrm{ind}}(H), \tag{2}$$

$$c_{\mathrm{ind}}(\mathcal{C}(H)) - \frac{|V(\mathcal{C}(H))|}{2} = b_{\mathrm{ind}}(H) - (m - n). \tag{3}$$

Proof. Take any good set in the completion and intersect it with $V(H)$, obtaining S. No attachment bridge can lie in the selected subgraph, so every selected vertex of H has exactly two selected neighbours within H. Thus S is good in H.

If $v \in S$, every root attached to v is unselected, and each corresponding copy of P contributes at most six vertices. If $v \notin S$, each attached copy contributes at most seven. Conversely, these bounds are attainable simultaneously, for any fixed good S, by Lemma 2.1. There are no edges between different copies of P. The optimum extending S is therefore

$$\begin{aligned} |S| + 6\sum_{v\in S} d(v) + 7\sum_{v\notin S} d(v) &= 7D + \sum_{v\in S}(1 - d(v)) \\ &= 7D + \sum_{v\in S}(d_H(v) - 2) \\ &= 7D + |S \cap V_3(H)|. \end{aligned}$$

The last equality uses $d_H(v) \geq 2$ for every vertex of a good set. Maximizing proves (2). Subtracting $(n + 13D)/2$ and using $D = 3n - 2m$ proves (3). □

Corollary 3.2 (Equivalent formulation)**.** *The cubic conjecture (1) holds if and only if every subcubic graph H satisfies*

$$b_{\mathrm{ind}}(H) \geq |E(H)| - |V(H)|. \tag{4}$$

Proof. Apply (1) to $\mathcal{C}(H)$ and use (3) for the forward implication. For the reverse implication take H cubic: then $b_{\mathrm{ind}}(H) = c_{\mathrm{ind}}(H)$ and $m - n = n/2$. □

For a subcubic graph of minimum degree at least two, the right side of (4) is $|V_3(H)|/2$. Thus in that case the statement asks for induced cycles covering at least half the degree-three vertices; degree-two vertices are counted with weight zero. For general subcubic H, the form $m - n$ also accounts for vertices of degree zero or one.

The reduction is quantitative. If H violates (4) by an integer $q > 0$, its completion violates (1) by exactly q, and has at most $40n$ vertices. This supplies a certified route from a prospective subcubic counterexample to a cubic one. It does not establish that either kind of counterexample exists.

The deficit also has an exact interpretation in the complement. This clarifies why a decomposition into induced cycles and a pseudoforest would be sufficient, but would require an additional existence argument.

Lemma 3.3 (Complement identity)**.** *If S is good in a subcubic graph H and $T = V(H) \setminus S$, then*

$$|S \cap V_3(H)| - (m - n) = |T| - |E(H[T])|.$$

Consequently, (4) is equivalent to the existence of a good set whose complement has at most as many edges as vertices. A sufficient, potentially stronger condition is that the complement be a pseudoforest: every component has at most one cycle.

Proof. Each selected degree-three vertex has exactly one neighbour outside S, and each selected degree-two vertex has none. Thus the cut between S and T has $|S \cap V_3(H)|$ edges. Since $|E(H[S])| = |S|$, counting all edges and vertices gives the identity. For a pseudoforest, each component has at most as many edges as vertices, so the right side is nonnegative. □

4 Reductions for a smallest counterexample

The branch formulation admits reductions that preserve the direction of a counterexample. They do not remove the cubic case.

Lemma 4.1 (Pruning and suppression)**.** *If a counterexample to (4) exists, one of minimum order is connected and has minimum degree at least two. Its degree-two vertices lie in pairwise vertex-disjoint triangles, each containing two degree-three vertices.*

Proof. Both b_{ind} and $m - n$ are additive over components, so at least one component of a counterexample is a counterexample.

If v has degree zero or one, put $H' = H - v$. Every good set of H' is good in H. Moreover, every vertex of degree three in H' has degree three in H, so $b_{\mathrm{ind}}(H') \le b_{\mathrm{ind}}(H)$. If v is a leaf, then $|E(H')| - |V(H')| = m - n$; if v is isolated, this difference is $m - n + 1$. Thus deleting v preserves a violation.

Now suppose w has degree two and its neighbours u, v are nonadjacent. Suppress w: remove w and its two incident edges and add uv. The resulting simple subcubic graph H' has one fewer vertex and one fewer edge, and its other vertex degrees are unchanged. Given a good set S' in H', include w precisely when both u, v belong to S'. This replaces a selected edge by a selected path, or leaves an unselected subdivision vertex. In either case it gives a good set of H with the same number of degree-three vertices. Hence $b_{\mathrm{ind}}(H') \le b_{\mathrm{ind}}(H)$, while $m - n$ is unchanged. Again a violation descends to a smaller graph.

We next exclude a pendant cycle C: all its vertices have degree two except a single degree-three vertex u, whose external neighbour is z. Remove $V(C)$ to obtain H'. The quantity $m - n$ decreases by one. If a good set S' in H' contains z, then z has degree two in H' and three in H, so the same set gains one counted branch vertex when regarded in H. If $z \notin S'$, adjoin the whole cycle C, gaining the branch vertex u. In both cases $b_{\mathrm{ind}}(H) \ge b_{\mathrm{ind}}(H') + 1$, so a counterexample again descends.

Every degree-two vertex is therefore in a triangle whose other vertices both have degree three: a triangle with two degree-two vertices would be pendant, and a triangle with three would be an entire component satisfying equality in (4). Finally, two distinct such triangles cannot share just one vertex, as that would require degree at least four. If they share an edge, its two degree-three endpoints and the two degree-two vertices form an entire diamond component. That component has $m - n = 1$ and a good triangle containing both degree-three vertices, so is not a counterexample. Thus the triangles are pairwise vertex-disjoint. □

More generally, subdividing edges cannot decrease b_{ind}: lift a good set by selecting an entire replacement path exactly when its original edge has both endpoints selected. Subdivision also preserves $m - n$. Consequently, subdivisions of a graph already satisfying (4) cannot give a counterexample. Suppressing a degree-two vertex in a triangle would introduce a parallel edge; the simplicity hypothesis prevents applying the preceding reduction there without an additional argument.

Lemma 4.2 (Consecutive degree-two triangles)**.** *Suppose a simple subcubic graph H contains two vertex-disjoint triangles abw and cdz. Assume w, z have degree two, a, b, c, d have degree three, and the remaining edges incident with these six vertices are precisely ua, bc, dv, where u, v are distinct vertices outside the triangles and are nonadjacent. Delete the six triangle vertices and add uv, obtaining H'. Then*

$$b_{\mathrm{ind}}(H) \geq b_{\mathrm{ind}}(H') + 2, \qquad |E(H)| - |V(H)| = |E(H')| - |V(H')| + 2.$$

Consequently, a smallest counterexample to (4) has no such configuration.

Proof. The assumptions ensure that H' is simple and subcubic; all surviving vertex degrees are unchanged. Let S' be any good set of H'. If $u, v \in S'$, replace its selected edge uv by the induced path $uabcdv$, adjoining a, b, c, d and leaving w, z unselected. This gains four selected branch vertices. If $u \in S'$ and $v \notin S'$, adjoin the triangle cdz. If $u \notin S'$ and $v \in S'$, adjoin abw. If neither endpoint belongs to S', again adjoin abw. In each of these last three cases the added triangle has no selected neighbour outside itself, and it gains exactly two branch vertices. These constructions preserve goodness in every case.

Deleting the configuration removes six vertices and nine edges; adding uv restores one edge. Thus $m - n$ decreases by two. The two inequalities show that a counterexample would descend to the smaller graph H'. The distinctness and nonadjacency assumptions are essential to this reduction as stated; the lemma does not dispose of its loop or parallel-edge alternatives. □

Lemma 4.3 (Deleting a good set and its neighbours)**.** *Let S be good in a simple subcubic graph H, let $T = N_H(S) \setminus S$, and put $H' = H - (S \cup T)$. Then*

$$b_{\mathrm{ind}}(H) \geq b_{\mathrm{ind}}(H') + |S \cap V_3(H)|,$$
$$(|E(H)| - |V(H)|) - (|E(H')| - |V(H')|) = \sum_{v \in T} (d_H(v) - 1) - |E(H[T])|.$$

Consequently, if S is nonempty and its branch count is at least the second expression, a violation of (4) descends to the smaller graph H'.

Proof. The union of S with any good set of H' is good in H, because all neighbours of S outside S were removed. Surviving vertices can only gain branch weight when viewed in H. Since $H[S]$ has $|S|$ edges, the number of deleted edges is $|S| + \sum_{v \in T} d_H(v) - |E(H[T])|$. Subtract the number $|S| + |T|$ of deleted vertices to obtain the identity. □

Lemma 4.4 (One expanded parallel edge)**.** *Suppose H contains vertices u, v, a, b, w with internal edges uv, ua, ab, bv, aw, bw. Assume w has degree two; the other four vertices have degree three; and their only external edges are ux, vy, where x, y have degree three. This configuration cannot occur in a minimum-order counterexample to (4).*

Proof. First suppose $x \neq y$ and $xy \notin E(H)$. Delete the five configuration vertices and add xy, obtaining a simple subcubic H'. The quantity $|E| - |V|$ decreases by two and all surviving degrees

are unchanged. If a good set of H' contains both x, y, replace its edge xy by the path $xuvy$, gaining two branch vertices. Otherwise adjoin the triangle abw, also gaining two.

If $x \neq y$ and $xy \in E(H)$, delete the five vertices without adding an edge. Now $|E|-|V|$ decreases by three. If neither x nor y is selected in a good set of H', adjoin the induced four-cycle $uabvu$, gaining four branch vertices. Otherwise adjoin abw, gaining two, and recover at least one further branch weight at a selected member of $\{x, y\}$, whose degree was reduced from three to two. Each lift therefore gains at least three.

Finally, if $x = y$, delete the configuration and x together. They have eight internal edges and one external edge, so $|E| - |V|$ decreases by three. Adjoin $uabvu$ to any good set of the remainder, gaining four branch vertices. All three cases make a counterexample descend to a smaller graph. □

Theorem 4.5 (A simple base for a smallest obstruction)**.** *If* (4) *fails, every minimum-order counterexample has the following form. Start with a connected simple cubic graph G of order n, choose a set A of fewer than $n/2$ edges, and replace each $uv \in A$ by ua, ab, bv, aw, bw, using three fresh vertices. Thus there is at most one triangle on each base edge. The case $A = \varnothing$ is allowed.*

Proof. By Lemma 4.1, the graph H is connected and has minimum degree two. Its degree-two vertices belong to disjoint triangles, each with two degree-three vertices. Call vertices outside these triangles base vertices. If there are none, the degree-three vertices themselves induce a 2-regular graph and give $b_{\mathrm{ind}}(H) = |V_3(H)| \geq |E(H)| - |V(H)|$, a contradiction.

Every triangle has exactly two external incidences. Following them partitions the triangles into chains between base vertices. Connectedness rules out a separate circular chain. Replacing each chain by an edge gives a connected cubic multigraph B. Ordinary edges between base vertices are chains of length zero. We next exclude loops and parallel edges in B.

A loop has a chain of $k \geq 1$ triangles, since H is simple. Its branch vertices form an induced cycle C of length $2k + 1$. Outside C are its k degree-two triangle tips and one further neighbour. The expression in Lemma 4.3 is at most $k + 2$, whereas C has $2k + 1 \geq k + 2$ branch vertices. The lemma excludes this loop.

For parallel edges, choose two chains between base vertices u, v. If a length-zero chain exists, include it in the pair. This ensures that the cycle following the two chains has no chord, including when there is a third parallel chain. Let K be the total number of triangles on the chosen chains. Simplicity gives $K \geq 1$. The cycle has $2K + 2$ branch vertices; its external neighbourhood consists of K degree-two tips and at most two further vertices. The deletion expression is at most $K + 4$. If $K \geq 2$, the cycle has at least that many branch vertices, contradicting minimality. If $K = 1$, the two chains are one ordinary edge and one triangle, exactly the configuration of Lemma 4.4. Its external neighbours have degree three: a base vertex cannot be adjacent to a degree-two triangle tip. That lemma gives the contradiction.

Hence B is simple. A chain of length at least two contains a pair to which Lemma 4.2 applies. If the chain has length two, the outside endpoints are nonadjacent because B has no parallel edge; for a longer chain, the next triangle endpoint has no edge back to the first base vertex. Thus every chain has length zero or one, as claimed.

Write $k = |A|$. The union of the k gadget triangles is good, since different triangles are separated by base vertices, and has $2k$ branch vertices. Also $|E(H)| - |V(H)| = n/2 + k$. If $k \geq n/2$ this union already proves the required bound. Therefore $k < n/2$. □

This is a necessary form for a smallest obstruction, not its exclusion. In particular, the unrestricted simple cubic case remains when $A = \varnothing$.

Proposition 4.6 (Sparsity of the expanded edges)**.** *Let H_A be a minimum-order counterexample in the form supplied by Theorem 4.5. For every nonempty $B \subseteq A$, put $U = V(B)$ and $q_B = |E(G[U])\setminus A|$. Then*

$$2|B| + q_B \leq 2|U| - 1. \tag{5}$$

In particular, A is a forest. Every nontrivial component of this forest induces in G a tree or a unicyclic graph.

Proof. Let S be the union of the gadget triangles for B. These triangles have no edges between them, so S is good and contains $2|B|$ branch vertices. Its outside neighbourhood in H_A is exactly U, consisting of base vertices of degree three. The edges of $H_A[U]$ are precisely $E(G[U]) \setminus A$. Lemma 4.3 therefore deletes $S \cup U$ at cost $2|U| - q_B$ in $|E| - |V|$, while its good-set lift gains at least $2|B|$ branch vertices. Minimality forces $2|B| < 2|U| - q_B$, proving (5).

If A contained a cycle, take its edge set as B. Then $|B| = |U|$, contradicting the inequality. Hence A is a forest. For the edge set of any nontrivial component, $|U| = |B| + 1$, so $q_B \leq 1$. The induced graph on this component's vertices is its marked tree together with at most one further edge. □

The entire collection of subset inequalities has an equivalent description on a smaller auxiliary multigraph. This observation does not require H_A to be a counterexample.

Lemma 4.7 (Contracting the marked forest)**.** *For a simple cubic graph G and $A \subseteq E(G)$, condition (5) for every nonempty $B \subseteq A$ is equivalent to the following two conditions. First, A is a forest. Second, form a multigraph J whose vertices are the nontrivial components of that forest, with one edge for each unmarked edge of G having both endpoints in $V(A)$, between their components. Retain loops and parallel edges. Then*

$$|E(J[W])| \leq 2|W| - 1 \qquad \text{for every nonempty } W \subseteq V(J).$$

A loop counts as one edge in this inequality. The empty marking satisfies both descriptions vacuously.

Proof. The cycle argument above proves necessity of the forest condition without using minimality. If W is a collection of its components, take B to contain all their edges. Then $|V(B)| - |B| = |W|$ and $q_B = |E(J[W])|$, giving necessity of the displayed inequality.

Conversely, take nonempty $B \subseteq A$, let c be the number of components of $(V(B), B)$, and let W consist of the components of A that meet B. Since A is a forest, $|V(B)| = |B| + c$ and $c \geq |W|$. Every edge counted by q_B is among the edges of $J[W]$. Hence $q_B \leq 2|W| - 1 \leq 2c - 1$, which is (5). □

The forest restriction does not assert that A is a matching. Nor do these conditions exclude $A = \varnothing$: they leave the unrestricted cubic case unresolved.

The following exact calculation records a separate constraint on a possible obstruction built across a bridge. It helps distinguish rooted bounds from the unrooted conjecture.

Lemma 4.8 (Doubling a rooted attachment)**.** *Let Q have one vertex r of degree two and all other vertices of degree three. Put $q = |V(Q)|$, and let A and B be the maximum orders of good sets containing and avoiding r, respectively. Use $A = -\infty$ if no good set contains r. Join the roots of two disjoint copies of Q by an edge to form $D(Q)$. Then q is odd, $D(Q)$ is cubic, and*

$$c_{\mathrm{ind}}(D(Q)) = \max\{2B, A + B\}.$$

In particular, $D(Q)$ is a counterexample if and only if $2B < q$ and $A + B < q$.

Proof. The degree sum of Q is $3q-1$, so q is odd. The added edge is a bridge and cannot occur in a selected cycle. At most one of its endpoints can therefore be selected. The two possible optimum cardinalities are $2B$ and $A+B$. Each finite candidate value is attained by the corresponding choices in the two copies. □

For example, if Q is obtained by subdividing an edge e of a cubic graph G, then $B = c_{\mathrm{ind}}(G-e)$ and $A = 1+\max\{|S| : S \text{ is good in } G,\ e \subseteq S\}$, with the same infeasibility convention. Notice that deleting e can create new good sets, so $c_{\mathrm{ind}}(G-e)$ cannot simply be replaced by $c_{\mathrm{ind}}(G)$ in this test.

Theorem 4.9 (Bridge symmetrization)**.** *Let G be a connected cubic graph with a bridge. Deleting the bridge gives rooted graphs Q_1, Q_2, with roots at its endpoints. Write $q_i = |V(Q_i)|$, and let A_i, B_i be their root-in and root-out good-set optima. Define*

$$\delta_i = \max\{0, A_i - B_i\}, \qquad d_i = c_{\mathrm{ind}}(D(Q_i)) = 2B_i + \delta_i,$$

where an infeasible root-in state has $A_i = -\infty$ and $\delta_i = 0$. Then

$$c_{\mathrm{ind}}(G) = \frac{d_1 + d_2 + |\delta_1 - \delta_2|}{2}. \tag{6}$$

Consequently, if G violates (1), then at least one of $D(Q_1), D(Q_2)$ also violates it.

Proof. An induced union of cycles cannot use the bridge, so at most one root is selected. Each allowable pair of local good sets gives a global good set. Thus

$$c_{\mathrm{ind}}(G) = \max\{B_1+B_2, A_1+B_2, B_1+A_2\} = B_1 + B_2 + \max\{\delta_1, \delta_2\}.$$

Substitute $2\max\{\delta_1,\delta_2\} = \delta_1+\delta_2+|\delta_1-\delta_2|$ and apply Lemma 4.8. If both $d_i \geq q_i$, equation (6) gives $c_{\mathrm{ind}}(G) \geq (q_1+q_2)/2$. Taking the contrapositive proves the last assertion. Each $D(Q_i)$ is a simple cubic graph of order $2q_i$. □

Thus searching for a counterexample with a bridge can be restricted to symmetric joins without losing existence. This is not an order-preserving reduction: the new counterexample can be larger than G. It also places no restriction on a bridgeless counterexample. In particular, it does not itself prove the original conjecture.

Proposition 4.10 (One edge replaced by a triangle)**.** *Let G be cubic of order $n = 2h$, and let $e = uv$. Replace uv by the edges ua, ab, bv, aw, bw, where a, b, w are new vertices, obtaining a subcubic graph H. Define*

$$B_e = c_{\mathrm{ind}}(G-e), \qquad T_e = \max\{|S| : S \text{ good in } G,\ u,v \in S\},$$

and $C_e = \max\{|S| : S \text{ good in } G,\ u,v \notin S\}$. As usual an infeasible maximum is $-\infty$. Then

$$b_{\mathrm{ind}}(H) = \max\{B_e, T_e+2, C_e+2\}, \qquad |E(H)| - |V(H)| = h+1. \tag{7}$$

If $c_{\mathrm{ind}}(G) \geq h$ but H violates the branch inequality, then

$$c_{\mathrm{ind}}(G) = h, \qquad B_e \leq h, \qquad T_e, C_e \leq h-2.$$

In particular every maximum good set of G contains exactly one endpoint of e. Subdividing e once to obtain Q then gives a cubic counterexample $D(Q)$ of order $2n+2$ with $c_{\mathrm{ind}}(D(Q)) \leq n$.

Proof. Consider a good set of H. If it contains w, it must contain a, b, and it must avoid u, v. The triangle contributes two branch vertices and the remaining vertices form a good set of G avoiding both endpoints. If w is unselected, either both a, b are unselected, or both are selected. The former case is exactly a good set of $G - e$. In the latter case u, v must be selected, and contracting the path $uabv$ to uv gives a good set of G containing both endpoints. Conversely, each of these three kinds of good set extends as described. This proves the maximum formula. The operation adds three vertices and four edges, proving the second identity.

If the branch inequality fails, integrality in (7) gives $B_e \le h$ and $T_e, C_e \le h - 2$. Every good set of G either contains both endpoints and has size at most T_e, or remains good after deleting e and has size at most B_e. Thus $c_{\mathrm{ind}}(G) \le h$; the hypothesis gives equality. A maximum good set cannot contain both endpoints or neither endpoint, by the bounds on T_e, C_e.

For the subdivided graph Q, the two optima in Lemma 4.8 are $A = T_e + 1 \le h - 1$ and $B = B_e \le h$. Thus $A + B \le n - 1$ and $2B \le n$. Lemma 4.8 gives $c_{\mathrm{ind}}(D(Q)) \le n$, below its half bound $n + 1$. □

Proposition 4.11 (Simultaneous replacements)**.** *Let G be a simple cubic graph of order n, let $A \subseteq E(G)$, and perform the edge-triangle replacement of Proposition 4.10 on each edge of A, using disjoint new vertices. Denote the resulting subcubic graph by H_A. For an Eulerian edge set $F \subseteq E(G)$, let $R = V(F)$ be its support, put $X = E(G[R]) \setminus F$, and let $\delta(R)$ be the edge boundary of R. Include $F = \varnothing$. Then*

$$b_{\mathrm{ind}}(H_A) = \max_{\substack{F \\ \text{Eulerian} \\ X \subseteq A}} \{|R| + 2(|A| - |A \cap \delta(R)| - |X|)\}. \tag{8}$$

In particular, a connected base requires only $2^{n/2+1}$ Eulerian states, independently of the number of expanded edges.

Proof. In a cubic graph an Eulerian edge set has degree zero or two at every vertex. Thus X is a matching: each vertex of its support already has two incident edges in F.

Project a good set of H_A onto the original vertices, obtaining R. An expanded edge has precisely the three local possibilities described in Proposition 4.10: an empty gadget, the path through its two branch vertices, or its whole triangle. Let F consist of the unexpanded edges with both endpoints selected and the expanded edges whose path is selected. Every selected original vertex has degree two in F, so F is Eulerian and has support R. Every edge of X must have been expanded and must use an empty gadget. A marked edge crossing $\delta(R)$ likewise has an empty gadget. The remaining marked edges can contribute two branch vertices apiece, giving the upper bound in (8).

Conversely, take any Eulerian F with $X \subseteq A$. Select R; on marked edges of F select the two branch vertices, and on marked edges outside R select the entire triangle. Leave the other gadgets empty. Every selected degree is two, and the branch count is exactly the displayed expression. This proves equality. The state count is the dimension formula for the binary cycle space. □

Theorem 4.12 (A fractional cut certificate)**.** *Let G be a simple cubic graph of order n. Suppose a probability distribution on good sets has edge-cut probabilities $p_e = \Pr(e \in \delta(S))$ satisfying*

$$\sum_{e \in E(G)} \min\{p_e, 1 - p_e\} \ge n/2. \tag{9}$$

Then $b_{\mathrm{ind}}(H_A) \ge n/2 + |A|$ for every $A \subseteq E(G)$. Consequently every cubic completion $\mathcal{C}(H_A)$ satisfies the original conjecture. More quantitatively, if $\tau = \sum_e \min\{p_e, 1 - p_e\}$, then $b_{\mathrm{ind}}(H_A) \ge |A| + \lceil \tau \rceil$.

Proof. Every good set S of G lifts to a good set of H_A with branch count $|S| + 2|A \setminus \delta(S)|$. Since G is cubic, each selected vertex has exactly one neighbour outside S, so $|\delta(S)| = |S|$. The expected surplus of this lift over $n/2 + |A|$ is therefore

$$\begin{aligned}\mathbb{E}|S| - n/2 + |A| - 2\sum_{e\in A} p_e &= -n/2 + \sum_{e\notin A} p_e + \sum_{e\in A}(1-p_e)\\ &\geq -n/2 + \sum_{e} \min\{p_e, 1-p_e\} \geq 0.\end{aligned}$$

Some lift has nonnegative surplus. Also $|E(H_A)| - |V(H_A)| = n/2 + |A|$, so Theorem 3.1 gives the assertion for cubic completions. The same calculation bounds the expected branch count below by $|A| + \tau$; some lift attains at least this expectation, and rounding up gives the quantitative assertion. In particular, $\tau > n/2$ gives a strict unit surplus. Reversing pairs of triangle replacements preserves this surplus for arbitrary chain lengths. □

The hypothesis is sufficient, not necessary. For the sharp twenty-six-vertex graph formed from two copies of P joined at their roots, it cannot hold. If p is the probability that the joining bridge crosses the selected cut, Lemma 2.1 gives $\mathbb{E}|S| \leq 12 + p$: sets selecting neither root have size at most twelve, and those selecting one have size at most thirteen. Therefore every distribution satisfies

$$\sum_{e} \min\{p_e, 1-p_e\} \leq 12 + p - \max\{0, 2p-1\} \leq 25/2 < 13.$$

The graph itself satisfies the original conjecture with equality. This explains why the certificate cannot settle the unrestricted problem by itself.

Theorem 4.13 (Conditional cut certificates)**.** *Let G be simple cubic of order n, and let $P_0, Q_0 \subseteq E(G)$ be disjoint sets of forced marked and forced unmarked edges. Consider a probability distribution on Eulerian edge sets F such that every chord set $X = E(G[V(F)]) \setminus F$ is contained in P_0. Write $p_e = \Pr(e \in \delta(V(F)))$. If*

$$\sum_{e\in P_0}(1-p_e) + \sum_{e\in Q_0} p_e + \sum_{e\notin P_0\cup Q_0} \min\{p_e, 1-p_e\} \geq n/2, \tag{10}$$

then every marking $P_0 \subseteq A \subseteq E(G) \setminus Q_0$ satisfies $b_{\mathrm{ind}}(H_A) \geq n/2 + |A|$.

Proof. Put $R = V(F)$ and $C = \delta(R)$. Counting the unused incidence at each vertex of R gives $|R| = 2|X| + |C|$. Thus the surplus of the lift in Proposition 4.11 is exactly

$$|R| + 2(|A| - |A \cap C| - |X|) - (n/2 + |A|) = |A \triangle C| - n/2. \tag{11}$$

Every state in the distribution is feasible because $X \subseteq P_0 \subseteq A$. The expected symmetric-difference size is $\sum_{e\in A}(1-p_e) + \sum_{e\notin A} p_e$, bounded below by the left side of (10). Hence some feasible lift has the desired surplus. □

The earlier criterion is the case $P_0 = Q_0 = \varnothing$. Unlike it, the conditional version may use Eulerian supports with chords, provided those chords are forced to be expanded. For the sharp double of P, split on the joining bridge e: use $(P_0, Q_0) = (\varnothing, \{e\})$ or $(\{e\}, \varnothing)$. Exact rational distributions with ten states each attain thirteen in both cases. These two certificates cover every subset of its thirty-nine edges. They are checked without an optimization package in Section 14.4; a solver status is not used as proof.

Corollary 4.14 (Arbitrary triangle chains)**.** *Suppose G has a distribution satisfying (9), or has conditional certificates whose covered markings together include every $A \subseteq E(G)$. Replace each edge uv by any number $k_e \geq 0$ of triangles in series: for $k_e > 0$, use triangles $a_i b_i w_i$ with connecting edges $ua_1, b_1a_2, \ldots, b_{k_e-1}a_{k_e}, b_{k_e}v$. Use distinct new vertices for different chains. Let $K = \sum_e k_e$ and let H be the resulting subcubic graph. Then*

$$b_{\mathrm{ind}}(H) \geq n/2 + K, \qquad |V(\mathcal{C}(H))| = n + 16K,$$

and $\mathcal{C}(H)$ satisfies the original half bound.

Proof. Repeatedly remove two consecutive triangles using Lemma 4.2. The outside endpoints are distinct and nonadjacent because the base is simple and each replacement is a chain with disjoint internal vertices. This leaves one triangle on the edges with odd k_e and none on the others. Theorem 4.12 or Theorem 4.13 applies to this reduced graph. Reversing each reduction restores two units of branch count and two units of $|E| - |V|$. Hence $b_{\mathrm{ind}}(H) \geq n/2 + K$. The K vertices w_i are the only degree-two vertices, so the completion formula gives order $n + 3K + 13K = n + 16K$ and the half bound. □

The fixed attachment also admits certificates retaining its root state. Use labels $\ell_i^j = 4i + j$, $r_i^j = 4i + 2 + j$, and root 12. The following two lists each contain four good sets with six branch vertices:

root avoided	$\{4,6,7,8,9,10\}$, $\{5,6,7,8,9,11\}$,
	$\{0,1,2,8,10,11\}$, $\{0,1,2,9,10,11\}$
root selected	$\{1,2,3,4,5,6,12\}$, $\{0,2,3,4,5,7,12\}$,
	$\{0,2,4,6,8,10,12\}$, $\{1,2,4,7,9,11,12\}$.

In either uniform distribution every edge crosses at most two of the four selected cuts. These claims follow directly by checking the nineteen edges of P.

Proposition 4.15 (Preserving both attachment states)**.** *Replace the edges of P by triangle chains with total length K. The resulting subcubic graph has, separately with its root selected and avoided, a good set containing at least $6 + K$ branch vertices. Attach a fresh copy of P at each of the K new degree-two tips, leaving the original root of degree two. The resulting rooted graph Q has order $13 + 16K$ and has root-in and root-out optima satisfying*

$$A(Q) \geq 7 + 8K, \qquad B(Q) \geq 6 + 8K.$$

Proof. First expand a subset A of edges just once. For either of the displayed four-set distributions, the ordinary good-set lift has expected branch count $6 + 2\sum_{e \in A}(1 - p_e) \geq 6 + |A|$, since $p_e \leq 1/2$. Its original root state is unchanged. Remove pairs of consecutive triangles to handle arbitrary chains, and reverse the reductions as in Corollary 4.14. The lifts retain the root state and give branch count at least $6 + K$.

For any such good set, completing a new tip contributes seven from its attached cap if the tip is absent, or six plus the selected tip if it is present. The combined contribution is seven in either case. Thus the completed set has size equal to its branch count, plus $7K$, plus the indicator that the original root is selected. The asserted lower bounds follow. They are not claims of equality for the expanded attachment. □

5 Sharp graphs with bridges

Let T be a tree in which each vertex has degree one or three, and let t be its number of degree-three vertices. Replace each such vertex by a triangle, using its three vertices as distinct attachment points for the three incident edges of T. Replace each leaf by a copy of P, using its root as the attachment point. Retain the edges corresponding to T as bridges. Write G_T for the resulting graph. There are $t+2$ leaves, so

$$|V(G_T)| = 3t + 13(t+2) = 16t + 26.$$

Theorem 5.1 (Exact equality family)**.** *For every such tree,*

$$c_{\mathrm{ind}}(G_T) = 8t + 13 = \frac{|V(G_T)|}{2}.$$

In addition, $V(G_T)$ partitions into two good sets of this size.

Proof. For $t = 0$, the tree is an edge joining two leaves. The graph consists of two copies of P joined at their roots. At most one root is selected, so the optimum is $7 + 6 = 13$. The complementary sets from Lemma 2.1, with opposite choices in the two copies, give the required partition.

Now let $t \geq 1$. The subgraph T_0 induced by the nonleaves of T is a tree on t vertices. Because attachment edges are bridges, a good set selects either an entire internal triangle or none of it. The selected triangles form an independent set X of T_0. If v has ℓ_v leaf neighbours in T, selecting its triangle gains three vertices and reduces the available contribution from each attached leaf copy from seven to six. For a fixed X, the optimum is therefore

$$7(t+2) + \sum_{v \in X} (3 - \ell_v) = 7(t+2) + \sum_{v \in X} d_{T_0}(v).$$

For any independent X, the last sum counts each edge of T_0 at most once, so it is at most $t-1$. Either bipartition class attains $t-1$. This proves the maximum $7(t+2) + t - 1 = 8t + 13$.

For the partition statement, assign the triangles according to a bipartition of T_0. At every leaf copy, assign its root-containing seven-set to the class opposite its neighbouring triangle, and its complementary six-set to the triangle's class. The two resulting sets partition all vertices and are both good: every attachment edge joins opposite classes. The preceding degree count shows that both sets have $8t + 13$ vertices. □

These orders are 2 modulo 4, so these graphs have no square factor. The theorem thus provides sharp examples outside the class treated in the preceding note. The partition property is a familiar sufficient condition for half-coverage: in a cubic graph, a partition into two induced 2-regular subgraphs has a perfect matching between its parts. The equality upper bound here follows from the attachment calculation, not merely from existence of that partition.

Corollary 5.2 (Decorated bridge trees)**.** *In any graph G_T, replace any of its bridges and any edges inside its leaf copies of P by arbitrary triangle chains. Let K be the total number of new triangles, and attach a fresh copy of P at each new degree-two tip. The resulting cubic graph has order $16t + 26 + 16K$ and induced cycles vertex number at least $8t + 13 + 8K$.*

Proof. Expanding a bridge by one triangle and completing its tip subdivides the corresponding tree edge by a new degree-three tree vertex, with a new leaf attached there. Repeating this handles all bridge chains and produces another tree of the same type. An expanded original leaf cap with K_i new triangles is the rooted graph of Proposition 4.15. In either root state it contributes at least

$8K_i$ more vertices than the original cap. Use a half-size good set from Theorem 5.1 on the tree after bridge expansions, and replace the states in its expanded leaf caps by these witnesses. Root states, and hence validity across bridges, are preserved. Each operation adds sixteen vertices and at least eight selected vertices, giving the assertion. Internal edges of the original degree-three-vertex triangles are not among the edges allowed by this corollary. □

6 An exact edge replacement

This section permits loops and parallel edges in an auxiliary base multigraph B; a loop contributes two to its vertex degree. The constructed graphs remain simple. Let $\nu_2(B)$ be the maximum number of vertices covered by an ordinary 2-regular subgraph of B. Inducedness is not required, and loops and pairs of parallel edges are allowed as cycles in this auxiliary parameter.

Put $R = F - e$, where F and e are as in Section 2. The endpoints $a = r_0^0$ and $b = \ell_1^0$ are the two attachment vertices of R. For a selected set in R, every selected nonattachment vertex must have internal selected degree two. At an attachment use state 0 for unselected, state 1 for selected with internal degree two, and state 2 for selected with internal degree one. State 2 requires a selected neighbour across the future attachment edge.

Lemma 6.1 (Exact attachment table)**.** *The only feasible pairs of attachment states in R are $(0,0),(0,1),(1,0),(2,2)$. For each of these four pairs, the maximum number of selected vertices is six.*

Proof. An odd number of state-2 attachments is impossible by the degree sum formula. For states with no 2, the selected set is good in $F - e$, giving the upper bound six by Lemma 2.1. For $(2,2)$, adding e gives a good set in F, with the same bound.

State $(1,1)$ is impossible. Selecting a forces both ℓ_0 vertices to be selected; selecting b similarly forces both r_1 vertices. Neither Q_0 nor Q_1 can be full, since the square analysis in Lemma 2.1 would then empty the other squares. Thus they are exactly these three-sets. Their doubled sides require both sides of Q_2 to be selected. This makes Q_2 full, again contradicting that analysis.

For explicit attainments, number $\ell_i^0, \ell_i^1, r_i^0, r_i^1$ as $4i, 4i+1, 4i+2, 4i+3$, respectively, so $(a,b) = (2,4)$. The following table gives selected six-sets:

$(0,0)$	$1,3,5,7,9,11$
$(0,1)$	$4,6,7,8,9,10$
$(1,0)$	$0,1,2,8,10,11$
$(2,2)$	$0,2,4,6,8,10.$

Their internal degrees verify the stated attachment states. □

For each edge uv of B, take a fresh copy of R, remove uv, and join u to its attachment a and v to its attachment b. For a loop, both joins use the same base vertex but distinct attachment vertices. Call the result $\mathcal{I}(B)$.

Theorem 6.2 (Cycle-packing encoding)**.** *If B is a cubic multigraph on n vertices, then $\mathcal{I}(B)$ is simple and cubic, and*

$$|V(\mathcal{I}(B))| = 19n, \qquad c_{\mathrm{ind}}(\mathcal{I}(B)) = 9n + \nu_2(B).$$

Proof. There are $3n/2$ edge copies, each adding twelve vertices. Every attachment receives exactly one new edge, and every base vertex retains its three incidences. Distinct copies have disjoint vertices, so even loops and parallel edges in B create no loops or parallel edges in $\mathcal{I}(B)$.

In a good set of $\mathcal{I}(B)$, mark a base edge precisely when its copy of R has state $(2,2)$. Lemma 6.1 shows that these are exactly the copies whose attachment edges occur in selected cycles. Each selected base vertex has exactly two marked incidences; an unselected base vertex has none. The marked edges therefore form an ordinary 2-regular subgraph of B, covering at most $\nu_2(B)$ base vertices. Each edge copy contributes at most six, proving the upper bound $6(3n/2) + \nu_2(B)$.

Conversely, choose a maximum ordinary cycle packing in B and select its base vertices. Use the displayed $(2,2)$ six-set in each marked edge copy and the $(0,0)$ six-set in every other copy. All selected vertices have degree two, and the resulting good set has the asserted cardinality. □

For completeness, the elementary bound needed to apply this formula also holds for multigraphs with loops.

Lemma 6.3. *Every connected cubic multigraph B on n vertices satisfies $\nu_2(B) \geq n/2 + 1$. If B is bridgeless, then $\nu_2(B) = n$.*

Proof. Choose a maximum ordinary cycle packing, covering a set S of s vertices. Its complement T induces a forest, since any cycle there could be added to the packing. Let c be the number of components of this forest, and let q count the edges of $B[S]$ outside the packing. Thus $|E(B[S])| = s + q$, the cut has $s - 2q$ edges, and $|E(B[T])| = n - s - c$. Counting all $3n/2$ edges gives

$$3n/2 = (s+q) + (s-2q) + (n-s-c), \qquad s = n/2 + q + c.$$

If T is nonempty, then $c \geq 1$. Otherwise $s = n$ and $q = n/2 \geq 1$. This proves the first bound.

For a bridgeless cubic multigraph, a loop would force its remaining incident edge to be a bridge. The graph is therefore loopless, and Petersen's perfect-matching theorem [4] gives a spanning 2-factor, proving $\nu_2(B) = n$. □

Consequently every connected B satisfies

$$c_{\mathrm{ind}}(\mathcal{I}(B)) \geq \frac{|V(\mathcal{I}(B))|}{2} + 1.$$

For bridgeless B, the exact value is $c_{\mathrm{ind}}(\mathcal{I}(B)) = 10|V(\mathcal{I}(B))|/19$. The additive constant one is sharp: take a tree with t vertices of degree three and $t+2$ leaves, and put a loop at each leaf. Only those loops can occur in an ordinary cycle packing, so $n = 2t+2$ and $\nu_2(B) = t+2$. The resulting simple cubic graph has order $38t + 38$ and induced cycles vertex number $19t + 20$.

7 A criterion for nonuniform edge replacements

The preceding exact table is unusually simple. The following sufficient condition applies to different fragments on different edges and explains why a larger collection of replacements cannot produce a counterexample over a bridgeless base.

Let B be a loopless bridgeless cubic multigraph on n vertices. For each edge e, let R_e be a simple graph with two designated degree-two attachments and all other degrees three. Replace e by R_e as before, and call the resulting simple cubic graph G. For each feasible attachment state (i,j), define

$$f_e(i,j) = \max_S \{2|S| - |V(R_e)|\},$$

where the maximum is over locally valid selected sets in that state; put $f_e(i,j) = -\infty$ for an infeasible state. Use the states of Lemma 6.1, and write

$$C_e = \max_{i,j \in \{0,1\}} f_e(i,j), \qquad D_e = f_e(0,0), \qquad A_e = f_e(2,2).$$

Both C_e and D_e are finite, since the empty set is allowed.

Theorem 7.1 (Matching criterion)**.** *Suppose every A_e is finite. If*

$$\sum_{e\in E(B)} (3C_e + D_e + 2A_e) \geq 0, \tag{12}$$

then $c_{\mathrm{ind}}(G) \geq |V(G)|/2$. In particular, it suffices that $3C_e + D_e + 2A_e \geq 0$ for each edge separately.

Proof. First leave every base vertex unselected. In each fragment use a closed state attaining C_e. These choices are compatible and give a good set with doubled surplus

$$Z_0 = \sum_e C_e - n.$$

Next, for a perfect matching M of B, select every base vertex. Use state $(0,0)$ in fragments on M and state $(2,2)$ in all other fragments. Each base vertex then has exactly two selected neighbours, and all other selected degrees are two by the local states. This good set has doubled surplus

$$Z_M = n + \sum_{e\in M} D_e + \sum_{e\notin M} A_e.$$

There is a probability distribution on perfect matchings of B under which each edge occurs with probability $1/3$. Indeed, the constant edge vector $1/3$ has degree sum one at every vertex. Every odd vertex set has an odd cut, and bridgelessness makes that cut have at least three edges. The vector therefore satisfies the perfect-matching polytope inequalities; Edmonds' theorem [3] expresses it as a convex combination of perfect matchings. Consequently,

$$\mathbb{E}Z_M = n + \frac{1}{3}\sum_e (D_e + 2A_e), \qquad \frac{Z_0 + \mathbb{E}Z_M}{2} = \frac{1}{6}\sum_e (3C_e + D_e + 2A_e) \geq 0.$$

At least one of the constructed good sets has nonnegative doubled surplus, which proves the claim. □

If B has a proper three-edge-colouring, its three colour classes provide the required distribution directly, so this case uses no polyhedral result. Of the 27 retained two-attachment profiles described below, 21 have a finite $(2,2)$ state. Independent subset enumeration verifies $3C + D + 2A \geq 0$ for each of these 21 fragments. The theorem therefore applies to every choice and orientation of these fragments on every loopless bridgeless cubic base, with no bound on its order. This is an application to a specified fragment library; we have not proved the local inequality for arbitrary fragments, and the hypothesis on the base excludes bridges.

8 Two defects and a finite bipartite census

The following reduction isolates a restricted class of possible counterexamples. A set D is called 1-*limited dominating* if it dominates its complement and every vertex of D has at most one neighbour outside D. In a cubic graph the latter condition means that $G[D]$ has minimum degree at least two. This parameter and its relation to induced cycles are studied in [7].

Lemma 8.1 (Two-defect reduction)**.** *Let G be cubic of order n, and let D be a 1-limited dominating set of size $n/2+1$. Either D is good, or its complement S is good and $G[D]$ has exactly two vertices of degree three. In the second case, $G[D]$ consists of a theta graph or two cycles joined by a path, together with zero or more cycle components. Here a theta graph consists of three internally disjoint paths with common endpoints.*

Proof. Write $t = |D|$, $s = |S| = t - 2$, and let q count the vertices of degree three in $G[D]$. Its other vertices have degree two, so the cut has $t - q$ edges. Domination gives $t - q \geq s$, whence $q \leq 2$. The handshaking lemma makes q even. If $q = 0$, D is good. If $q = 2$, the cut has exactly s edges, so every vertex of S has exactly one neighbour in D and $G[S]$ is 2-regular.

The two degree-three vertices belong to the same component, since every component has an even number of odd-degree vertices. Suppress all degree-two vertices in that component, temporarily allowing loops and parallel edges. The resulting cubic multigraph on two vertices has either three parallel edges, or one edge joining the vertices and a loop at each. Expanding the suppressed paths gives the asserted theta or joined-cycle structure. Every remaining component is a cycle. □

For a bipartite G in the second case, the two degree-three vertices lie in opposite bipartition classes. To see this, write t_L, t_R for the numbers of vertices of D in the two classes and a, b for their numbers of degree-three vertices in $G[D]$. Counting internal edges in the two classes gives $2t_L + a = 2t_R + b$. The good set S is a union of even cycles, so its two classes have equal size. The cut is a matching saturating S and all the degree-two vertices of $G[D]$, giving $t_L - a = t_R - b$. These equations imply $a = b = 1$ and $t_L = t_R$. Consequently all three theta paths have odd length; in the other case the two cycles are even and the joining path has odd length. All additional cycle components are even.

Proposition 8.2 (A parity restriction)**.** *If $n \equiv 0 \pmod 4$, a cubic bipartite graph of order n has no 1-limited dominating set of size $n/2 + 1$. If $n \equiv 2 \pmod 4$, it has no such set of size $n/2$.*

Proof. In the first case, both $n/2 + 1$ and $n/2 - 1$ are odd, whereas Lemma 8.1 makes one of the corresponding sets a union of even cycles. In the second case, a dominating set D of size $n/2$ has at most $|D|$ cut edges and must dominate $n/2$ outside vertices. Equality forces every vertex of D to have internal degree two. Its odd cardinality is impossible in a bipartite graph. □

We exhaustively generated the second case of Lemma 8.1 for bipartite graphs of orders $10, 14, 18, 22, 26$. A template specifies the even cycle lengths of S, the core lengths just described, and the even cycle lengths of the remaining components of D. For each template we enumerate all colour-respecting matchings between S and the degree-two vertices of D. For a single S-cycle, rotation allows one specified attachment to be fixed; no such reduction is made when S has several components. If $|S| = 2h$, the numbers of tested matchings per template are respectively $(h-1)!h!$ and $(h!)^2$.

Fixing a bipartition for each template does not omit a gluing: a global interchange fixes the core's colour convention, and rotation by one vertex interchanges the colours of any even cycle component. Repeated isomorphic completions are permitted. The counts therefore refer to labelled completions, not distinct isomorphism classes.

Order	Templates	Tested completions
10	1	2
14	3	36
18	12	4,320
22	24	207,360
26	88	36,115,200

Every completion admitted a good set of size at least $n/2 + 1$. The search constructs an explicit set and directly checks its cardinality and every selected degree before accepting a case. Witnesses

are checked during the run; the saved output consists of aggregate counts and rolling digests, not a separate certificate for every completion. Thus this is a reproducible finite computation.

The program `verify_two_defect_search.py` independently checks the structural list against all 22 complements of order 14 generated by `geng -qb -d2 -D3 14 15:15`, and all four cycle factors of order 12 generated with degrees fixed at two. It also compares the new counting solver against direct subset enumeration on all 112 connected cubic graphs through order 12. All checks pass. The source programs, templates, outputs and scope statements are retained in the accompanying research record.

As a consequence of this computation and the preceding lemmas, every cubic bipartite graph of order at most 26 that has a 1-limited dominating set of size at most $n/2+1$ satisfies (1). Indeed, counting the cut first gives $|D| \geq n/2$; at equality D itself is good. A set of size $n/2+1$ is handled by the parity restriction, the good-set alternative, or the enumeration. At order six the second alternative cannot occur, since it would require a good set of order two. There is no proof here that an arbitrary cubic graph has such a dominating set, nor does the finite census establish the two-defect case at unbounded order.

9 Obstructions to exchange arguments

An exchange argument would begin with a good set S and replace it by a larger good set. Define its *deletion radius*

$$\rho_G(S) = \min\{|S \setminus T| : T \text{ is good and } |T| > |S|\},$$

with value $+\infty$ if no improvement exists. The next statements identify limitations of two possible exchange strategies. They are not counterexamples to the induced cycles conjecture.

Lemma 9.1 (Girth forces large exchanges)**.** *Let G be a simple cubic graph of girth g. Let S be good, suppose its edge boundary is a matching, and let q count the vertices outside S with no neighbour in S. Every good set $T \neq S$ satisfies*

$$g \leq 6|S \setminus T| + q.$$

In particular, $\rho_G(S) \geq \lceil (g-q)/6 \rceil$.

Proof. Put $R = S \setminus T$, $A = T \setminus S$, and let B be the vertices of $S \cap T$ whose two selected incident edges change. Write $r = |R|$, $a = |A|$, and $b = |B|$. Each vertex of B loses one neighbour in R and gains one in A: two distinct pairs of edges at a cubic vertex have exactly one common edge. There are at most $2r$ edges from R to $S \setminus R$, so $b \leq 2r$.

Every vertex of A, except possibly the q vertices counted in the hypothesis, has a neighbour in S. That neighbour must belong to $R \cup B$. A retained vertex outside B already has its two selected neighbours and cannot acquire another. Since the boundary is a matching, these neighbours in $R \cup B$ are distinct. Hence $a \leq r + b + q$.

The symmetric difference of the edge sets of $G[S]$ and $G[T]$ has degree two at precisely the vertices of $R \cup A \cup B$, and degree zero elsewhere. It is nonempty because $S \neq T$, so it contains a cycle. Therefore

$$g \leq r + a + b \leq 2r + 2b + q \leq 6r + q.$$

□

Theorem 9.2 (No uniform bounded exchange)**.** *For every nonnegative integer r, there is a connected simple cubic bipartite graph G with a good set S of size $|V(G)|/2 - 1$ such that $G - S$ has minimum degree two and every distinct good set T satisfies $|S \setminus T| > r$.*

Proof. We first construct such two-defect graphs with arbitrarily large girth. Fix an integer $g \geq 3$ and put $k = 2^g$. Take a cycle S of length $6k$, and a disjoint theta graph D whose three paths each have length $2k+1$. Thus $|D| = 6k+2$; its two branch vertices have degree three and all others have degree two. Its branch vertices are in opposite bipartition classes. Each class of S and each class of the degree-two vertices of D has size $3k$. Join these vertices by arbitrary colour-respecting perfect matchings. Every resulting graph is simple, cubic, bipartite and connected, with S good and exactly two defects outside S.

Among these finitely many completions, choose one minimizing lexicographically the numbers of cycles of lengths $3, 4, \ldots, g-1$. Suppose its girth is $\ell < g$. The fixed cycle and theta have girths $6k$ and $4k+2$, respectively, so a shortest cycle uses a matching edge uv, with $u \in S$. The union of the radius-$(g-1)$ balls around u and v has at most

$$2(1 + 3(2^{g-1} - 1)) = 3 \cdot 2^g - 4$$

vertices. There are $3k = 3 \cdot 2^g$ matching edges in the same colour class as uv. Since these edges form a matching, one of them, say xy with $x \in S$, has both endpoints outside those balls. Replace uv, xy by uy, xv.

This switch preserves the template, simplicity and bipartiteness. A new cycle using just one new edge has length greater than g, since its remaining path joins the two separated endpoint sets. A new cycle using both new edges splits into two old paths. If these paths join u to x and v to y, both have length at least g. Otherwise they join u to v and x to y, avoiding the deleted edges; each has length at least $\ell - 1$, by the old girth. In this case the new cycle has length at least $2\ell > \ell$. Thus no cycle of length at most ℓ is created, while a cycle of length ℓ is destroyed. This contradicts the lexicographic choice. The chosen completion has girth at least g.

Now take $g = 6r+3$. Lemma 9.1, with $q = 2$, gives $6r+3 \leq 6|S \setminus T| + 2$ for every distinct good set T, and hence $|S \setminus T| > r$. □

The theorem disproves the assertion that a fixed number of deletions always suffices to improve a two-defect set. It does not decide whether a larger good set exists in every constructed graph. The distinction matters: large girth alone supplies an obstruction to the proposed local step, not to the original inequality.

There is also a barrier to a different move with no length bound. Call a replacement of a good set S by a good set T a *single-cycle exchange* when $E(G[S]) \triangle E(G[T])$ is the edge set of one simple cycle. Every difference of two good sets in a cubic graph is a disjoint union of cycles, but performing those cycles one at a time need not preserve inducedness or the objective value.

Theorem 9.3 (Traps for single-cycle improvement)**.** *For every integer $t \geq 2$, there is a connected simple cubic graph G_t of order $16t + 26$ and a good set S_t such that*

$$|S_t| = 7t + 14, \qquad c_{\mathrm{ind}}(G_t) = 8t + 13,$$

and no sequence of nondecreasing single-cycle exchanges starting at S_t reaches a larger good set. For $t = 2$, the exact loss barrier is two: every single-cycle-exchange sequence reaching a set larger than 28 visits a set of size at most 26, and a sequence with sizes $28, 27, 26, 29$ exists.

Proof. Use the sharp bridge construction of Section 5, taking the internal vertices of its underlying tree to form a path on t vertices. There are t triangles and $t+2$ copies of the cap P. Leave every triangle unselected and choose the seven-vertex root-containing set in every cap. The resulting good set S_t has size $7(t+2)$, whereas the exact optimum from Section 5 is $8t + 13$. Their difference is $t - 1 > 0$.

Every edge joining different pieces is a bridge. Thus a simple cycle, and hence a single-cycle exchange, lies wholly within one cap or one triangle. Each cap already contributes its maximum of seven vertices. Changing its root from selected to unselected would leave at most six, so a nondecreasing sequence cannot make that change. Every triangle has at least one adjacent leaf cap, because the internal tree is a path. Selecting the triangle while that cap's root is selected would put both ends of a bridge in a good set, which is impossible. Neutral cap moves therefore cannot unlock any triangle, and no nondecreasing sequence reaches an improvement.

For $t = 2$, each triangle is adjacent to two caps. An improvement must select a triangle, since the four caps together contribute at most 28. Immediately before the first triangle is selected, both adjacent cap roots must be unselected. The total size at that point is at most $6 + 6 + 7 + 7 = 26$. This bound is attainable. With the numbering of Section 2, replace the root-containing set $\{0, 2, 4, 6, 8, 10, 12\}$ in either adjacent cap by $\{0, 3, 5, 6, 8, 10\}$. The latter is good and excludes the root; the edge-set difference is a single cycle of length seven. Doing this in both adjacent caps and then selecting their triangle gives the stated sizes $28, 27, 26, 29$. □

The accompanying verifier checks all 107 simple cycles of P and thus all 430 simple cycles of the 58-vertex example. Exactly 88 flips yield good sets; none improves the initial size 28. The escape sequence is checked directly. It also independently verifies minimum deletion radii for 600 seeded two-defect instances of orders 26, 38 and 50, exhausting 48,480 candidates with smaller deletion counts. Those sampled radii are one or two; their small values do not extend to the unbounded-girth family.

10 Odd spanning factors and exact edge profiles

An *odd spanning factor* of a cubic graph G is a spanning subgraph M in which each vertex has degree one or three. An *isolated edge* means a component isomorphic to K_2. The following elementary reformulation separates parity from the inducedness condition.

Proposition 10.1. *Good sets of a cubic graph G are in bijection with odd spanning factors having no isolated-edge component. The correspondence is*

$$S \longmapsto M_S = (V(G), E(G) \setminus E(G[S])), \qquad M \longmapsto L(M),$$

where $L(M)$ is the set of degree-one vertices of M. If $n = |V(G)|$, $k(M)$ is the number of components, and $\beta(M) = |E(M)| - n + k(M)$, then

$$|L(M)| - \frac{n}{2} = n - |E(M)| = k(M) - \beta(M). \tag{13}$$

Consequently, the conjecture is equivalent to the existence of an odd spanning factor with no isolated edge and at most n edges.

Proof. For a good S, vertices in S have degree one in M_S, and all others have degree three. An edge joining two degree-one vertices in M_S would be an edge of $G[S]$, contrary to its deletion. Conversely, every edge outside an odd factor M has both endpoints in $L(M)$, since a degree-three vertex uses all its incident edges. Each leaf therefore has two neighbours in $L(M)$ outside M. An additional edge inside $L(M)$ would be an isolated edge of M. In its absence, $G[L(M)]$ is 2-regular, and the two maps are inverse. Writing $\ell = |L(M)|$, the degree sum gives $2|E(M)| = \ell + 3(n - \ell) = 3n - 2\ell$; this proves the identities. □

Corollary 10.2. *If a cubic graph has an odd spanning pseudoforest with no isolated edge, it satisfies the half bound. More precisely, the leaves form a good set of size $n/2 + t$, where t is the number of tree components.*

Proof. A tree component contributes 1 to $k - \beta$, and a unicyclic component contributes zero. Apply (13). □

This sufficient condition is precisely the earlier pseudoforest complement condition in another form: M_S consists of $G - S$ with one pendant edge for each vertex of S. It is not a proved existence theorem for arbitrary cubic graphs. Scott's perfect-forest theorem, for which Gutin gives a short proof [2], supplies an odd spanning forest in every connected graph of even order, but permits isolated-edge components. It therefore does not supply the missing condition. Moreover, replacing "pseudoforest" by "forest" cannot work universally: the graph F of Section 2 has $c_{\mathrm{ind}}(F) = 6$, whereas an isolated-edge-free odd forest on its twelve vertices would give at least seven leaves by Corollary 10.2.

Proposition 10.3. *Let G be connected and cubic of order n, and let $\mathcal{O}(G)$ be its odd spanning factors. Then $|\mathcal{O}(G)| = 2^{n/2+1}$. For $M \in \mathcal{O}(G)$ write $J(M)$ for its isolated-edge components, regarded as a set of edges. For each edge $e = uv$, put*

$$B_e = c_{\mathrm{ind}}(G - e), \qquad T_e = \max\{|S| : S \text{ good in } G,\ u, v \in S\},$$

with $T_e = -\infty$ when no such set exists. Then

$$B_e = \max\{|L(M)| : M \in \mathcal{O}(G),\ e \in E(M),\ J(M) \subseteq \{e\}\},$$
$$T_e = \max\{|L(M)| : M \in \mathcal{O}(G),\ e \notin E(M),\ J(M) = \varnothing\}.$$

All these optima and the complete good-set size distribution can be computed in $O(n2^{n/2+1})$ time and polynomial working space.

Proof. Symmetric difference with $E(G)$ identifies odd factors with the binary cycle space. A spanning tree gives a fundamental cycle basis of dimension $|E(G)| - n + 1 = n/2 + 1$. Thus all factors are obtained exactly once by toggling subsets of this basis.

For any odd factor, a leaf has exactly two neighbours joined to it by edges outside the factor; both are leaves. The only extra edges of $G[L(M)]$ are therefore the edges in $J(M)$. Accordingly, $(G - e)[L(M)]$ is 2-regular exactly when $e \in E(M)$ and $J(M) \subseteq \{e\}$. Conversely, every good set in $G - e$ gives such an odd factor by removing its induced edges from G. This proves the formula for B_e. If $J(M)$ is empty, the two endpoints of an edge e are selected together exactly when $e \notin E(M)$; this proves the second formula.

For each factor, compute its leaves and isolated edges in $O(n)$ time. If there are no isolated edges, update the histogram and all eligible edge optima. If there is exactly one, update the deletion optimum for that edge alone. Factors with two or more contribute to none of these quantities. A binary Gray order toggles one fundamental cycle per step; its length is at most n. The basis, current factor, histogram, optima, and witnesses require only polynomial space. □

Subdividing e gives a rooted fragment of order $n + 1$ with exact root states $A_e = 1 + T_e$ and B_e. By Lemma 4.8, joining two copies at their roots gives a counterexample precisely when

$$2B_e < n + 1 \quad \text{and} \quad B_e + T_e < n.$$

An impossible root-in state is omitted from the maximum. Thus the enumeration checks a complete family of rooted counterexample tests for each input graph.

The implementation `odd_factor_enumerate.cpp` and its standard-library verifier use different search spaces. Direct enumeration of all vertex subsets on all 112 connected cubic graphs through order twelve agrees in every histogram coefficient and all 1,899 edge profiles; it checks 369,040 subsets. A further seeded sample contains 240 labelled graphs of orders 24 through 42, including sharp constructions, one to eight degree-preserving switches from them, retained search candidates, and random cubic graphs. Exhausting 261,726,208 odd factors gives 11,526 exact edge profiles. Every input satisfies the original half bound, and every deletion has $B_e \geq n/2$; none passes the counterexample test above. Five inputs attain the original half bound. All maximum and edge-profile witnesses are retained and checked directly. These 240 inputs are a sample, not an exhaustive census up to order 42. Their upper bounds rely on complete cycle-space enumeration; they are not asserted on the strength of witnesses alone.

11 Repairing odd pseudoforests

For an odd spanning factor M, write $j(M) = |J(M)|$ for its number of isolated-edge components. A *cycle exchange* replaces its edge set by $E(M) \triangle E(C)$, where C is a simple cycle of G. The resulting factor is still odd. Here we permit intermediate factors with isolated edges, but require every intermediate factor to be a pseudoforest. This is a different state space from the good-set exchanges in Section 9.

The following first step has an unrestricted proof.

Lemma 11.1. *If P is a perfect matching of a nonempty finite simple cubic graph G, then G has an induced cycle that is not P-alternating.*

Proof. If G is not bipartite, a shortest odd cycle is induced and is not alternating. Suppose instead that G has bipartition (A, B). Orient matching edges from A to B, and all other edges from B to A. A cycle is directed exactly when it is P-alternating. If every induced cycle were alternating, this would be a cyclic orientation of G in the sense of Speyer [1]. Speyer's decomposition theorem implies that cyclically orientable graphs are 2-degenerate: their blocks are built from edges and cycles by successively attaching a cycle along one edge. In particular, each block is series-parallel, and the block decomposition preserves 2-degeneracy. Thus a nonempty such graph has a vertex of degree at most two, contrary to cubicity. □

Theorem 11.2 (A repair from every perfect matching)**.** *Every perfect matching P of a nonempty finite simple cubic graph admits a cycle exchange to an odd spanning pseudoforest M with $j(M) < j(P) = n/2$.*

Proof. Choose an induced nonalternating cycle C by Lemma 11.1. Put $r = |V(C)|$, $k = |E(C) \cap P|$, and $b = r - 2k > 0$. The b vertices of C not covered by $E(C) \cap P$ have distinct matching partners outside C, because C is induced. Toggle $E(C)$.

If $k = 0$, the changed component is C with one pendant edge at every vertex. It is unicyclic. If $k > 0$, deleting the k matching edges from C leaves disjoint paths; adding the unchanged matching edges at their b internal vertices attaches distinct pendant leaves. These components are trees. Unaffected components remain isolated edges. Thus the new factor is an odd spanning pseudoforest.

Exactly $k + b$ old matching components are affected. When $k > 0$, let a be the number of remaining cycle paths consisting of a single edge; these are precisely the new isolated edges, and $a \leq k$. Set $a = 0$ when $k = 0$. Consequently

$$j(M) = \frac{n}{2} - k - b + a \leq \frac{n}{2} - b < \frac{n}{2}.$$ □

This theorem establishes only the first step. After the exchange, nontrivial tree or unicyclic components may be present, so the matching hypothesis is lost. Reapplying the theorem is not justified. A proposed sufficient repair assertion for the full conjecture was:

> From every odd spanning pseudoforest with $j > 0$, there is a finite sequence of cycle exchanges through odd spanning pseudoforests such that j never increases and eventually strictly decreases.

Every connected cubic graph has even order and an odd spanning forest. Iterating the stated assertion would reach $j = 0$ and then prove the half bound by Corollary 10.2. The assertion is *false*, as proved in Theorem 13.2 below. We retain the earlier finite tests and first-step obstructions to make clear why they did not settle this stronger assertion.

Proposition 11.3 (Certified exchange obstructions)**.** *The following finite examples exist.*

1. *A connected cubic graph of order 14 has an odd spanning forest with one isolated edge from which no single cycle exchange both preserves the pseudoforest condition and reduces j. One neutral exchange followed by an improving exchange does succeed.*

2. *A connected cubic graph of order 16 has an odd spanning pseudoforest with 15 edges and one isolated edge, but every nontrivial cycle exchange preserving the pseudoforest condition has at most 14 edges.*

3. *The one-step obstruction in the first assertion also occurs in a connected bipartite cubic graph of order 36. Again, two exchanges suffice, the first neutral in j.*

None of these graphs is a counterexample to the induced cycles conjecture.

Finite certificate and verification. Number edges in graph6 order: increasing larger endpoint, then increasing smaller endpoint, starting at index zero. An integer mask selects exactly the edges with set bits. The first two instances have the following graph6 codes and factor masks:

14	`M¿@?aGqBCCoDOCW?`	1296836,
16	`O???C@_cSoAgQ_P_D_@D?`	4716667.

The first factor is three copies of $K_{1,3}$ and one K_2. The second has components with $(|V|, |E|)$ equal to $(6, 6), (8, 8), (2, 1)$.

For the third graph, take the bipartite double cover of the 18-vertex graph with graph6 code `Q????A?oE?kCEGL?DO?i?Ao?F??`. Label its two sheets $0, \ldots, 17$ and $18, \ldots, 35$; replace each edge uv by $u(v + 18)$ and $(u + 18)v$. Use factor mask 2812704630600655 in the resulting graph6 edge order. Its components have orders $10, 6, 6, 12, 2$; only the order-12 component is unicyclic, and the others are trees.

Direct vertex-path enumeration gives respectively 93, 196 and 11,976 simple cycles. Each cycle is toggled, and connected components of the resulting factor are inspected directly. In the 14-vertex case, the complete table of allowed moves is

$(\lvert E(M')\rvert, j(M'))$	$(7,7)$	$(8,5)$	$(9,3)$	$(10,1)$	$(11,2)$	$(12,1)$
number of cycles	4	20	32	15	1	2.

In particular j never drops below one in a single move. A two-move repair is the sequence of factor masks $1296836, 313647, 1930178$. In the 16-vertex case, the numbers of allowed moves grouped by new edge count $8, 9, 10, 11, 12, 13, 14$ are $2, 14, 32, 37, 35, 33, 13$; there are none with at least 15 edges.

In the 36-vertex case, every allowed move has $j \geq 1$; exactly 153 have $j = 1$. A two-move repair has masks

$$2812704630600655, \quad 2812730224997562, \quad 2812704640340943.$$

The last factors in the two repair sequences have no isolated edge and have respectively nine and 24 leaves, giving good sets. Direct subset enumeration gives $c_{\text{ind}} = 10$ for the 16-vertex graph. Thus all three satisfy the original half bound.

The standard-library program `verify_pseudoforest_repair.py` reproduces the cycle lists, full move tables, and shortest repairs. Its output retains all graph edges and factors explicitly, so the certificates do not require graph6 software. Component traversal and direct edge-parity recursion are independent of the C++ disjoint-set and cycle-space implementation. These are computer-checked finite statements, not an unrestricted repair theorem. □

The stronger assertion allowing neutral moves passed the following finite tests. Every odd spanning factor was enumerated for each input; for each pseudoforest with $j > 0$, all direct moves were examined until an improvement was found, or a breadth-first search explored moves with unchanged j until an improvement was reached.

Input collection	Graphs	Odd factors	Factors needing a neutral move
All connected cubic graphs, $n \leq 18$	45,982	44,513,544	214
All connected cubic graphs, $n = 20$	510,489	1,045,481,472	1,234
Seeded sample, $24 \leq n \leq 42$	240	261,726,208	144
Bipartite double covers, $28 \leq n \leq 36$	39	18,776,064	48

All tested factors needing neutral moves admitted a subsequent strict improvement. In the complete census through order twenty, each such factor needs exactly two exchanges: one neutral and one improving. At order twenty there are 920,048,096 odd pseudoforests, of which 889,866,061 contain isolated edges. The 1,234 factors requiring a neutral move occur in 291 distinct graphs. Completeness of the graph inputs up to isomorphism relies on `geng -c -d3 -D3 20`; the factor search exhausts each graph's cycle space. These counts concern the stronger repair assertion; the original conjecture was already checked through order twenty-two in the literature [6]. The last two collections are samples, not censuses of their orders. The double covers come from the 39 distinct graphs supporting the 214 small neutral-move cases. Their lifted initial factors all admit immediate improvements; the 48 neutral-move cases are different factors found by exhausting each cover's cycle space. The 36-vertex certificate above is one of these. A further 7,616 completions of its fixed factor, obtained by degree-preserving switches among the other edges, produced ten one-step obstructions, all of vertex connectivity two. Three-connectivity does not remove the obstruction, as the next finite certificate shows.

Proposition 11.4 (A three-connected one-step obstruction)**.** *There is a three-connected cubic graph G of order twenty and an odd spanning pseudoforest M with one isolated edge such that no single cycle exchange preserving the pseudoforest condition reduces $j(M)$. Two exchanges suffice. Nevertheless, $c_{\text{ind}}(G) = 14 > |V(G)|/2$.*

Finite certificate and verification. In the graph6 and edge-mask convention above, take

$$G = \texttt{S??????o@_@_gCGoWCAD?Cg?X??WO?IG?}, \qquad M = 983379956.$$

Deleting any one or two vertices leaves the graph connected. The factor M consists of the isolated edge $\{1, 14\}$ and an 18-vertex unicyclic component, whose unique cycle is the triangle on $\{5, 11, 19\}$.

Direct vertex-path enumeration gives 664 simple cycles. Toggling each and inspecting the resulting components gives no pseudoforest with $j = 0$, and exactly 57 with $j = 1$. For the neutral moves, the new edge counts $15, 16, 17, 18, 19$ occur respectively $9, 18, 19, 9, 2$ times. The complete move table is retained with the certificate.

A shortest repair has factor masks

$$983379956 \longrightarrow 983451959 \longrightarrow 977257976.$$

Their respective pairs $(|E(M)|, j(M))$ are $(19, 1), (17, 1), (19, 0)$. The last factor's leaves form the good set

$$\{0, 1, 2, 4, 7, 9, 10, 13, 14, 17, 18\}.$$

Independent enumeration of all 2^{20} vertex subsets gives the good-set size polynomial

$$\sum_{S \text{ good}} x^{|S|} = 1 + 2x^3 + 4x^4 + 2x^5 + 5x^6 + 9x^7 + 10x^8 + 13x^9 + 13x^{10} + 14x^{11} + 11x^{12} + 6x^{13} + x^{14}.$$

Thus the maximum is fourteen. The standard-library verifier `verify_round9.py` checks the connectivity, all cycles, the repair, and this polynomial independently of the C++ census. □

The census finds exactly three three-connected graphs supporting one-step obstructions at order twenty, with one such factor in each. The other 1,231 factors have vertex connectivity one or two (152 and 1,079 factors, respectively). Of the 1,234 factors, 1,218 have $j = 1$ and sixteen have $j = 2$; all supporting graphs have girth three. None of the 214 neutral cases through order eighteen is three-connected. Hence twenty is the minimum order of a three-connected one-step obstruction within the complete census. This is a computational minimality statement, not a general theorem about higher girth or larger orders. The independent verifier checks every one of the 1,448 reported neutral cases through order twenty, including the absence of a direct improvement and an explicit two-exchange repair. It does not independently repeat the entire 556,471-graph census.

A different shortcut is to minimize the global potential $h(M) = |E(M)| + j(M)$ over all odd spanning factors. For an odd pseudoforest, $h(M) \leq n$: indeed $|E(M)| = n - t(M)$, and every isolated edge is a tree component, so $j(M) \leq t(M)$. However, a minimum of h need not be attainable with $j = 0$. For the ten-vertex graph with graph6 code `I?BeeOwM?`, exhausting its 64 odd factors gives

$$\min_{j(M)>0} h(M) = 8 < 9 = \min_{j(M)=0} h(M).$$

A factor with mask 9700 attains the left side and consists of two copies of $K_{1,3}$ and one K_2. Direct enumeration of all 2^{10} vertex subsets gives $c_{\text{ind}} = 6$, so this example also satisfies the original conjecture. It rules out this particular global minimization argument and the stronger rounding claim $c_{\text{ind}}(G) \geq |L(M)| - j(M)$, whose right side here is seven.

Independently, direct edge-parity enumeration checks all 12,296 odd factors in the 112 connected cubic graphs through order 12. Of these, 10,911 are pseudoforests, and all 9,334 having an isolated edge admit an immediate improvement. It also checks 17,356 induced-cycle repairs from all 1,098 perfect matchings in these graphs against Theorem 11.2. The finite results do not bridge the gap from the proved first step to the unrestricted repair assertion, which is disproved in Theorem 13.2.

12 Single-edge replacement and repair simulation

The fragment $R = F - e$ from Section 6 provides a useful test of attempts to enlarge the repair obstructions. Its effect can be determined for every cubic base graph, without a size bound. Let H be a simple cubic graph, let $uv \in E(H)$, and form H_R by deleting uv, adding a disjoint copy of R, and joining u and v to its ports $a = 2$ and $b = 4$, respectively.

Theorem 12.1 (Exact single-edge replacement)**.** *With $B_{uv} = c_{\mathrm{ind}}(H - uv)$ and T_{uv} as in Section 10,*

$$c_{\mathrm{ind}}(H_R) = 6 + \max\{B_{uv}, T_{uv}\} = 6 + \max\{c_{\mathrm{ind}}(H - uv), c_{\mathrm{ind}}(H)\}.$$

In particular, replacement by R preserves the conjectured half bound whenever it holds for H.

Proof. Consider any good set in H_R. By Lemma 6.1, its restriction to R has at most six vertices and one of the boundary states $(0,0), (0,1), (1,0), (2,2)$. In any of the first three states, each selected vertex outside R has no selected neighbor in R: a port in state 1 is selected but already has two selected neighbors inside R, so its outside neighbor must be unselected. The outside restriction is therefore good in $H - uv$ and has size at most B_{uv}. In state $(2,2)$ both u and v must be selected. Replacing their edges to R by uv makes the outside restriction good in H, with both ends of uv selected. Its size is at most T_{uv}. This proves the upper bound.

Conversely, a largest good set of $H - uv$ extends by a six-vertex $(0,0)$ state in R. A good set counted by T_{uv} extends by a six-vertex $(2,2)$ state. Thus the upper bound is attained. Every good set in H either contains both ends of uv, or remains good after uv is deleted. Since $T_{uv} \le c_{\mathrm{ind}}(H)$, this also proves the second equality. Finally, $|V(H_R)| = |V(H)| + 12$, so the increase of at least six preserves the half bound. □

Consequently a smallest counterexample cannot contain an induced copy of R whose two outside neighbors are distinct and nonadjacent: smoothing that copy to one edge would give a smaller simple cubic graph to which Theorem 12.1 applies. The restrictions on the outside neighbors ensure that smoothing creates neither a loop nor a parallel edge.

We next show that a natural choice of factor inside R also preserves repair paths. All vertex labels below are those of F in Section 2. Define a spanning tree T of R by

$$\begin{aligned} E(T) = \{&(0,2), (1,2), (1,3), (1,11), (4,6), (4,7),\\ &(5,6), (6,8), (8,10), (8,11), (9,11)\}. \end{aligned}$$

The ports $2, 4$ have degree two in T; every other vertex has degree one or three. Let

$$P = (2,0,3,5,7,4), \qquad E(U) = E(T) \triangle E(P).$$

Then U is the six-cycle $(1,3,5,6,8,11,1)$ with one leaf attached at each cycle vertex. Its leaves are $0, 2, 4, 7, 9, 10$. In particular, U is an odd spanning unicyclic graph with no isolated edge.

For an odd spanning factor M of H, define its lift by

$$\widehat{M} = \begin{cases} (M - uv) \cup T \cup \{ua, vb\}, & uv \in E(M),\\ M \cup U, & uv \notin E(M). \end{cases}$$

Here copies of the fragment are understood to be disjoint from $V(H)$.

Theorem 12.2 (A lift preserving repair paths)**.** *If M is an odd spanning pseudoforest of H, then $\widehat{M}$ is an odd spanning pseudoforest of H_R, with*

$$j(\widehat{M}) = j(M) - \mathbf{1}_{\{uv \in J(M)\}}, \qquad |L(\widehat{M})| = |L(M)| + 6.$$

Every simple-cycle exchange in H lifts to one simple-cycle exchange in H_R. Suppose $uv \notin J(M_0)$ and M_0 has a repair of length ℓ through odd pseudoforests with j nonincreasing. Then $\widehat{M}_0$ has such a repair of length at most ℓ.

Proof. If $uv \in E(M)$, its replacement by the tree T and the two attachment edges preserves the cycle rank and connectedness of the component containing uv. All old vertex degrees are unchanged; the ports have degree three, and other new vertices have degree one or three. No new isolated edge is created. An old isolated edge disappears precisely when it is uv. If $uv \notin E(M)$, the new factor adds the separate unicyclic component U, and all old components are unchanged. Both constructions add exactly six leaves. This proves the displayed identities and the pseudoforest assertion.

Let C be a simple cycle in H. If $uv \notin E(C)$, use the same cycle in H_R. Otherwise concatenate the edge ua, the internal path P, and the edge bv in place of uv. The resulting cycle has edge set

$$E(\widehat{C}) = (E(C) \setminus \{uv\}) \cup E(P) \cup \{ua, bv\}.$$

It is a simple cycle. The identity $T \triangle U = P$ gives

$$\widehat{M \triangle C} = \widehat{M} \triangle \widehat{C}.$$

For a sequence through odd pseudoforests, all intermediate lifted factors are therefore pseudoforests.

Truncate the original repair at its first strict decrease, so $j(M_i) = j(M_0)$ before the last state. Initially $uv \notin J(M_0)$, and hence $j(\widehat{M}_0) = j(M_0)$. Along the lifted sequence, either uv first becomes isolated in some M_i, immediately giving $j(\widehat{M}_i) = j(M_0) - 1$, or the strict decrease in $j(M_i)$ is reached first. Before that first decrease the lifted values remain constant. Taking this prefix gives a repair of length at most ℓ. □

This theorem rules out a particular search strategy: inserting this factor pair along an initially nonisolated edge cannot turn a two-exchange repair into one requiring three exchanges. It does not say that every odd pseudoforest of H_R is such a lift, and supplies no repair when none is known in H.

The standard-library verifier `verify_round10.py` checks the explicit fragment, the factor identities, and the cycle lifts. It also checks the exact optimum formula by vertex-subset enumeration on all 24 single-edge replacements of connected cubic graphs through order six. A separate exploratory test retains 2,895 triangle expansions of the three order-20 obstructions, at orders 22 through 42, and 216 replacements by R at order 32. All admit repair witnesses of one or two moves; every retained witness is independently checked. These samples are not exhaustive at their orders.

The 216 factors occur in 54 distinct labelled graphs. Exhausting all 7,077,888 odd factors of these inputs gives 5,895,198 pseudoforests, of which 5,878,454 have isolated edges. Every one of the latter admits a repair within two exchanges; 1,838 require a neutral first move. Explicit repair paths for all 1,838 are independently verified. For one such factor in each of the 54 graphs, the verifier also independently exhausts every simple cycle, totaling 968,802 cycles, and confirms that two exchanges are necessary. This does not independently repeat the full factor census. The proof above, rather than these finite tests, establishes the unrestricted lifting statement.

13 Unbounded barriers to pseudoforest repair

The monotone repair assertion fails in graphs with bridges, despite all the finite checks in Section 11. This failure can be arbitrarily large. In this section, a *repair sequence* for M is a finite sequence $M = M_0, M_1, \dots, M_s$ of odd spanning pseudoforests in which consecutive factors differ by a simple cycle and $j(M_s) < j(M)$. Intermediate defect counts are allowed to rise. Define the *excess repair barrier* by

$$h_G(M) = \min_{(M_i)} \bigl(\max_i j(M_i) - j(M)\bigr),$$

with value $+\infty$ if there is no repair sequence. A monotone repair must have $h_G(M) = 0$; positive barriers rule it out.

Use the thirteen-vertex cap P from Section 2, with $\ell_i^j = 4i + j$, $r_i^j = 4i + 2 + j$, and root $r = 12$. Two of its good sets are

$$X = \{0, 3, 5, 6, 8, 10\}, \qquad Y = \{0, 2, 4, 6, 8, 10, 12\}.$$

Write $N_X = E(P) \setminus E(P[X])$ and define N_Y similarly. The subgraph N_X is connected and unicyclic, with root degree two and all other degrees odd. The subgraph N_Y has the root isolated and a unicyclic component on the other twelve vertices. Their edge-set difference is the seven-cycle

$$Q = (0, 2, 12, 4, 6, 5, 3, 0). \tag{14}$$

These descriptions follow directly from the edge list in Section 2; neither subgraph has an isolated-edge component.

For $k \geq 3$, construct B_k from two disjoint k-cycles by joining one designated vertex of each with a bridge. Attach a separate copy of P, by its root, to every other vertex of the two cycles. Thus there are $2k - 2$ caps. The resulting graph is connected, simple and cubic, with

$$n_k = |V(B_k)| = 2k + 13(2k - 2) = 28k - 26.$$

Define M_k to contain every bridge and a copy of N_X in every cap, and no edge of either k-cycle. It has one isolated edge, namely the bridge between the designated vertices. Its other $2k - 2$ components are unicyclic graphs of order fourteen, each obtained from N_X by adding its adjacent cycle vertex as a leaf. In particular, M_k is an odd spanning pseudoforest with $j(M_k) = 1$.

Lemma 13.1 (Cost of freeing a cap)**.** *Let N be a spanning pseudoforest of P in which the root has degree zero or two and every other vertex has degree one or three. Let $j_P(N)$ count its isolated-edge components and put*

$$w(N) = j_P(N) + \mathbf{1}_{\{\deg_N(r)=0\}}.$$

If the component of N containing the root is a tree, then $w(N) \geq 1$.

Proof. For root degree zero the second summand already equals one. Suppose the root has degree two and $j_P(N) = 0$. Let L be the set of nonroot vertices of degree one in N. Every edge outside N has both ends in L, because the root and all degree-three vertices use every edge available to them in P. Each vertex of L consequently has two neighbours in L outside N. There is no additional edge of N inside L, since it would be an isolated-edge component. Hence $P[L]$ is 2-regular and avoids the root, so $|L| \leq 6$ by Lemma 2.1.

The degree sum gives $|E(N)| = 19 - |L|$. If the root component were a tree, the pseudoforest would have at most twelve edges, giving $|L| \geq 7$, a contradiction. □

Theorem 13.2 (Exact unbounded repair barriers)**.** *For every $k \geq 3$,*

$$c_{\mathrm{ind}}(B_k) = \frac{n_k}{2}, \qquad h_{B_k}(M_k) = k - 1.$$

The shortest repair sequence for M_k, allowing increases in j, has exactly k exchanges. One such sequence has defect counts

$$1, 2, \ldots, k, 0.$$

Thus there is no monotone repair sequence, and no constant bounds the necessary temporary increase in j, even for starting factors with just one isolated edge.

Proof. Every simple cycle of B_k lies in one cap or is one of the two displayed k-cycles, since every edge between pieces is a bridge. In any good set, each displayed cycle is therefore either entirely selected or entirely unselected. Both cannot be selected, because their joining bridge would then lie in the induced 2-regular subgraph. If neither is selected, the caps contribute at most $14(k-1)$ vertices. If one is selected, its adjacent cap roots must be unselected. The cap bounds $A = 7$, $B = 6$ give

$$c_{\mathrm{ind}}(B_k) = \max\{14(k-1),\ k + 6(k-1) + 7(k-1)\} = 14k - 13 = \frac{n_k}{2}.$$

Both bounds are attainable by the specified cap states.

Consider any repair sequence starting at M_k. Until the first exchange on a displayed k-cycle, all its vertices have factor degree one and all bridges remain present. If N_i are the current restrictions to the caps, the isolated-edge count is exactly

$$j = 1 + \sum_{i=1}^{2k-2} w(N_i). \tag{15}$$

The initial isolated bridge persists throughout this period, so some displayed cycle must be exchanged before j can reach zero.

Immediately before that first exchange, each of the $k-1$ cap components attached to that cycle must be acyclic. Otherwise the exchange adds the displayed cycle to a component that already contains a cap cycle, violating the pseudoforest condition. Lemma 13.1 and (15) now give $j \geq 1 + (k-1) = k$. Moreover, each of those caps started with its root in a cyclic component and so must have undergone at least one exchange. Since one exchange cannot cross a bridge, at least $k-1$ cap exchanges and the displayed-cycle exchange are necessary.

For attainment, choose one displayed cycle. In each of its $k-1$ caps, toggle Q from (14), changing N_X to N_Y. Each move leaves a unicyclic component of order twelve and makes the cap's attachment edge an isolated edge. Thus all intermediate factors are odd pseudoforests, with $j = 1, 2, \ldots, k$. Now toggle the chosen displayed cycle. It forms a unicyclic component with its k adjacent factor leaves and absorbs all k isolated edges. Every remaining component is unicyclic, so $j = 0$. This uses exactly k exchanges and attains the stated barrier. □

The smallest member of this family has order 58. Its repair has j-values $1, 2, 3, 0$ and factor-edge counts $57, 56, 55, 58$. Its final good set has 29 vertices. It is a counterexample to the monotone repair assertion, *not* to the induced cycles conjecture. The existence of an isolated-edge-free odd spanning pseudoforest in every cubic graph remains unproved; refuting this particular way of reaching one does not refute that existence claim.

The independent program `verify_round11.py` exhausts all 2^{19} cap-edge subsets and all 2^{13} cap-vertex subsets. There are 128 cap parity states. Exactly twelve have $w = 0$ and are pseudoforests; their roots lie in cyclic components, and they form a connected graph under cap-cycle exchanges.

The program checks all $12 \cdot 107$ moves from these states. For B_3, it also checks all 430 simple cycles at M_3 and directly verifies the $12^4 = 20,736$ global states in the neutral component, including the 41,472 attempted exits obtained by flipping a displayed triangle. Every such exit violates the pseudoforest condition. Cap moves remain in this product or increase j, so this accounts for the entire neutral component. Explicit attaining sequences and good sets are checked for $3 \le k \le 12$, at orders 58 through 310.

These bridge obstructions should not be read as evidence of the same barrier in bridgeless graphs. For the canonical bipartite double covers of these ten examples, an independently checked pair of simple-cycle exchanges gives j-values $2, 1, 0$ and a good set of order $n/2+4$. Their orders range from 116 to 620; connectedness and the absence of bridges are also checked directly. These certificates concern the specified covers only.

14 Checks and the remaining obstacle

14.1 Direct tests of the original bound

For an edge uv of a cubic graph B, an *edge-triangle replacement* deletes uv and introduces three vertices a, b, w and the five edges ua, ab, bv, aw, bw. It adds two branch vertices and one degree-two vertex. We applied this operation to every unordered pair of distinct original edges in each connected simple cubic graph B of orders twelve, fourteen and sixteen generated by `geng`. The two edges may share an endpoint. For the resulting graph H, writing $n = |V(B)|$, we have

$$|V(H)| = n + 6, \qquad |E(H)| = \tfrac{3}{2}n + 8, \qquad |E(H)| - |V(H)| = \tfrac{n}{2} + 2.$$

Every construction admitted a good set containing at least $n/2 + 2$ branch vertices. The counts refer to indexed constructions; different entries may be isomorphic.

Base order	Base graphs	Edge-pair constructions	Completion order
12	85	13,005	44
14	509	106,890	46
16	4,060	1,120,560	48

There are exactly two missing incidences in each H, so its cubic completion has order $n + 32$. The exact completion formula proves the original half bound for all these completions.

The search enumerates the binary cycle space of H. In a subcubic graph an Eulerian edge set has degree zero or two at every vertex, so its support is a union of vertex-disjoint cycles. Checking whether that support induces a 2-regular graph discards precisely the supports with additional chords or edges between cycles. Conversely, every good set occurs as such a support. For a connected graph this gives an exhaustive search with 2^{m-n+1} edge states, although the implementation stops when a qualifying positive witness is found. The separate standard-library program `verify_round12.py` reconstructs all 1,240,455 graphs and checks every saved vertex-set certificate directly, without using the search algorithm or an optimization package. The claimed completeness of the base list up to isomorphism relies on `geng`.

A second search tested 1,595 distinct labelled cubic graphs obtained by degree-preserving edge changes from the sharp family in Section 13 and from rings of the fragment R. Their orders range from 36 to 114. Every graph has a directly verified half-bound witness, including one case resolved by retrying a time-limited query. Fourteen named cubic graphs, with orders between 8 and 126 and LCF descriptions taken from the SageMath source [8], also have verified witnesses strictly exceeding

half their orders. These are lower bounds on individual graphs, not certified maxima or exhaustive order censuses. The verifier reconstructs each LCF graph independently.

Motivated by Theorem 4.9, a further search made 20,000 degree-preserving edge-switch proposals on rooted fragments of orders $13, 25, 29, 37$, corresponding to symmetric cubic joins of orders $26, 50, 58, 74$. The proposals contain 14,088 distinct labelled rooted graphs; they are not an exhaustive census. Every proposal has a verified half-bound witness in its full doubled cubic graph. Of the 20,000 records, 19,845 terminate once a strictly positive surplus is certified; their recorded local values are lower bounds. The remaining 155 records have exact profiles, all attaining the half bound upon doubling. These comprise 95 distinct labelled rooted graphs. The standard-library verifier `verify_round13.py` independently checks every full cubic witness and rechecks all 95 exact profiles by vertex-assignment recursion, using degree constraints to propagate forced choices. The search itself uses cycle-space enumeration. No timeout or early positive exit is interpreted as an optimum or a counterexample.

Lemma 4.2 is checked separately on all 3,615 edge replacements in labelled subcubic graphs of orders two through five. All 7,984 good-set lifts have the claimed gain. The proof of the lemma, rather than this finite check, establishes its unrestricted statement. None of these searches or reductions proves (1) for arbitrary cubic graphs.

14.2 A strictness census and a further synthesis attempt

A separate pass over all 556,471 connected cubic graphs of orders four through twenty finds precisely one isomorphism class attaining the half bound: the twelve-vertex Franklin graph F. For each of the other 556,470 graphs, a saved good set has at least $n/2 + 1$ vertices. These are positive certificates, not claimed exact optima. For the single equality case, exhaustive vertex-subset enumeration confirms optimum six, twenty maximum good sets, and the fact that every edge belongs to at least one maximum good set. An explicit relabeling identifies the graph with F.

The verifier `verify_round14.py` reconstructs every graph, checks connectedness and cubic degree, verifies the saved set, and compares the certificate list with the two existing `geng` input lists row by row. As before, isomorphism coverage relies on `geng`; the witness checks are independent of the search. Proposition 4.10 then proves the branch inequality for every single edge-triangle replacement in these bases. A strict base already supplies the needed extra vertex, and in F the through-edge optimum is six for every edge. There are 16,539,825 indexed choices of a base graph and an edge; this is a consequence of the census and the proposition, rather than separate enumeration of all those expanded graphs. Their cubic completions also satisfy (1), by Theorem 3.1.

Finally, two bounded synthesis runs sought a twenty-five-vertex rooted graph with $(A, B) \leq (12, 12)$ or $(A, B) \leq (13, 11)$, respectively. Either profile would violate the half bound after doubling. The runs fix the root neighbours and permit deletion of at most six edges of a specified labelled seed, together with the same number of insertions. They examined thirteen and ten candidates before their wall-time limits. Independent cycle-space enumeration checks both optima for all twenty-three candidates and verifies a half-bound witness in every full fifty-vertex cubic double. Neither run establishes infeasibility in its restricted search region. The degree-propagation oracle and cycle-space checker are also cross-checked against all vertex subsets in all 843 labelled subcubic graphs through order five. The triangle formula is checked by full vertex-subset enumeration for all 387 edge expansions of cubic bases through order ten and of F.

14.3 Certificates for simultaneous replacements

Theorem 4.12 has exact rational certificates for a list of 2,249 connected cubic bases through order twenty and one twenty-four-vertex base. The former list was selected by a cycle-space search for bases with optimum at most $n/2 + 2$; the certificate verification does not depend on that selection criterion. For 1,477 of the 2,250 bases, a uniform distribution on the listed good sets suffices. For the other 773, linear programming proposed a distribution, which was converted to rational coefficients. The separate verifier checks every listed set, the sum and sign of the coefficients, and (9) using exact `Fraction` arithmetic. Numerical tolerances are not acceptance criteria.

Each certificate covers every subset of edges simultaneously, and Corollary 4.14 extends it to arbitrarily long chains. Thus these are certificates for infinite families of cubic graphs, rather than a claim about a fixed list of expanded orders. The twenty-six-vertex sharp bridge graph does not meet the criterion, consistently with the upper bound proved after Theorem 4.12; its recorded distribution attains $25/2$ exactly.

The implementation of (8) is independently cross-checked against all edge subsets of the eight connected cubic bases through order eight. Its optimum formula is then checked against every vertex subset of 551 expanded graphs: all marking subsets for K_4, and all marking subsets of size at most two for the remaining bases through order eight. The verifier constructs and degree-checks a half-bound witness in all 551 full cubic completions, whose orders are at most one hundred.

An earlier search in this continuation followed sequences of replacements that decrease or preserve branch surplus, to depth four. It produced 11,193 distinct labelled expanded graphs; every saved lower-bound witness is checked directly. This restricted search does not cover sequences that temporarily increase surplus. A separate simultaneous-marking constraint model reported infeasibility for 1,387 selected small bases before its wall-time limit and for two further tight bases of orders twenty-four and twenty-six. No formal solver proof files are retained; these statuses are exploratory records. The exact rational distributions, not these solver statuses, justify the infinite-family assertions. All new data and checks are recorded by `verify_round15.py`. The original conjecture remains unresolved.

14.4 Normal form and conditional certificates

The reductions in Theorem 4.5 settle the loop and parallel-edge alternatives left by Lemma 4.2. They are structural proofs for arbitrary orders. A standard-library verifier checks the neighbourhood identity and its lifts on all 12,911 labelled subcubic graphs on one through six vertices. It tests 17,925 nonempty good sets and 17,945 lifts. The short parallel-edge configuration is checked in 2,463 bounded outside environments, with 4,850 good-set lifts, covering equal, adjacent, and nonadjacent outside neighbours. A further 55 loop and parallel-chain configurations check the neighbourhood reduction with longer chains. These finite checks support the implementation of the proofs; they are not an enumeration of all prospective counterexamples.

For the sharp twenty-six-vertex double of P, two conditional cut certificates have ten rational terms each and cut lower bound thirteen. The verifier reconstructs every Eulerian edge set, support, chord set, and cut; checks positive coefficients summing to one; and checks (10) using exact fractions. The two cases partition all $2^{39} = 549{,}755{,}813{,}888$ indexed zero-or-one marking choices. This is a certificate argument, not individual enumeration. Corollary 4.14 then covers arbitrary chain lengths on this base, which the original unconditional cut criterion could not certify.

The four-set root certificates are checked directly on P. Both root states are also checked arithmetically for all 2^{19} marking patterns. For the 191 patterns with at most two marked edges, actual rooted graphs and full cubic doubles are constructed; every doubled graph has a directly

checked half-bound witness. Their orders are at most ninety. These are positive witnesses, not claimed optima for the expanded graphs.

The program `verify_round16.py` performs these checks without an optimization package; it saves counts and data hashes. The data directory retains the rational certificates. The original conjecture remains unresolved, including the unexpanded cubic case in Theorem 4.5.

14.5 Marked-forest restrictions and rooted search

The proofs of Proposition 4.6 and Lemma 4.7 hold without an order bound. For an implementation check, `verify_round17.py` compares the full subset inequalities with the contracted-forest criterion for all 21,568 markings of the eight connected cubic bases through order eight. Of these markings, 13,284 are forests and 4,276 satisfy the stronger sparsity condition. For markings of size at most three, the verifier constructs 10,351 neighbourhood deletions and checks all 45,390 resulting good-set lifts. These counts describe finite checks of the formulas, not a resolution of the remaining counterexample search.

Two further constraint searches sought rooted graphs Q of order 29 with $A(Q), B(Q) \leq 14$. Such a graph would yield a 58-vertex counterexample by doubling. The searches fix the two root neighbours and bound the number of deleted seed edges by six or four, with degree-preserving insertions. The radius-six run stopped on a query timeout after one candidate; the radius-four run reached its wall time limit after six candidates. Neither region is proved infeasible. Independently enumerating binary cycle-space states checks the exact rooted profiles of all seven candidates, and the verifier checks a half-bound witness on each full cubic double. None is a counterexample. Search records, exact verification counts, and hashes are retained with the source.

14.6 Direct descent and four further certificates

A direct search on cubic graphs ordered candidates first by their maximum good-set size and then by the number of good sets attaining that size. It used degree-preserving edge switches, occasional upward moves, and restarts. Accepted moves were evaluated exactly by binary cycle-space enumeration; early exits recorded only positive lower-bound witnesses. Twelve seeds, including vertex expansions and sharp graphs assembled from the established attachments, led to 30,000 switch proposals at orders $24, 26, 28, 30, 32, 36, 38, 42$. Together with 185 retained vertex expansions, the data contain 30,185 records and 28,733 distinct labelled graphs. All retained witnesses are checked directly. An independent vertex-assignment recursion verifies the optima and maximum-set counts of all 45 recorded best graphs. No counterexample was found. This is a bounded heuristic search, with no claim of exhaustive coverage at these orders.

Four selected graphs were then tested for simultaneous edge-triangle replacements. A constraint solver reported infeasibility for all four counterexample queries; no formal infeasibility proofs were retained. Independently of these statuses, explicit rational cut certificates establish stronger positive results:

Base order	Certificate terms	τ	$\lceil \tau - n/2 \rceil$
26	84	96/7	1
28	10	29/2	1
30	10	31/2	1
32	12	33/2	1

The first certificate is uniform on maximum good sets; the others use rational distributions proposed by linear programming. The verifier checks every selected degree, coefficient, probability sum, and

cut total using exact fractions. The quantitative form of Theorem 4.12 and the chain reduction show that every cubic completion of arbitrary triangle-chain replacements on each of these four bases has induced cycles vertex number at least half its order plus one. These are proved infinite-family bounds, not a proof of the unrestricted conjecture.

The programs `cubic_multiplicity_descent.cpp` and `verify_round18.py`, seed construction source, compressed search records, rational certificates, and exact verification hashes are retained with the manuscript. Non-best search records are independently certified only as positive witnesses, rather than independent optimum computations.

14.7 Exhaustive edge neighbourhoods of two sharp graphs

A direct counterexample search starts from the sharp 24- and 26-vertex seeds retained in `round20_data/seed24.g6` and `seed26.g6`. The forbidden property is a good set of size at least $n/2$; no auxiliary inequality is substituted for the original conjecture. A constraint-generation search adds a clause excluding every qualifying good set of each candidate. Its two runs produced nine and seven candidates and stopped at wall-time limits, with 1,496 and 1,616 accumulated clauses. These solver outcomes are inconclusive, not unsatisfiability certificates.

A separate enumeration completely checks the specified finite regions. For each seed G, it generates every labelled simple cubic H on the same vertices satisfying $|E(G) \setminus E(H)| \leq 4$. For each set of k deleted seed edges, pair its $2k$ endpoint incidences in every possible way. Reject loops, repeated new edges, and any new edge already belonging to the seed. This generates exactly the desired edge changes: cubicity ensures that the added edges pair precisely the endpoint incidences of the deleted edges. Different pairings can yield the same labelled graph when incidences share a vertex, so the following counts are reconnection records, not counts of distinct graphs. Disconnected candidates are included.

Deleted edges k	24-vertex seed	26-vertex seed
0	1	1
1	0	0
2	984	1,182
3	35,592	47,464
4	1,937,772	2,849,236
Total	1,974,349	2,897,883

Every record has a good set of size at least $n/2$. Most reuse a previously found witness; exact vertex-assignment search supplies 383 and 967 additional witnesses, respectively. Thus neither of these entire finite neighbourhoods contains a counterexample. This conclusion does not exclude graphs outside the two regions.

The implementation is `exploratory/edge_ball_enumerate.cpp`. The independent standard-library verifier `verify_round20.py` checks all 1,350 additional witnesses directly. It separately enumerates the cycle spaces of all sixteen constraint-search candidates and confirms their reported optima and qualifying-set counts. For a test of the reconnection generator itself, an independent degree-sequence recursion enumerates all 70 labelled cubic graphs on six vertices and all 19,355 on eight vertices. For one seed of each order, the radius-four regions agree exactly with the C++ generator: 54 and 1,147 distinct graphs, respectively. These tests validate the small cases and retained witnesses; they do not constitute an independent re-enumeration of the two large regions.

14.8 Earlier checks and limitations

The accompanying standard-library program `verify_bridge_reduction.py` enumerates the good sets of F, $F - e$, and P and checks their claimed maxima. It constructs the completing witnesses for all 843 labelled subcubic graphs on one through five vertices and verifies their degrees, cardinalities, and the surplus identity. It also checks thirteen sharp tree constructions, including orders from 26 through 186. These finite checks support the implementation; the proofs above establish the unrestricted statements.

The program `verify_edge_inflation.py` independently enumerates all 2^{12} subsets of R to verify the complete attachment table. It checks ordinary packing optima for twelve small auxiliary multigraphs, the corresponding induced-cycle witnesses after inflation, and eleven members of the new family attaining the additive constant one.

Further searches found no violation among 9,000 local changes to tight graphs of orders 24, 26, and 42; every optimization in those runs finished with a certified optimum from the solver. A floating-point weighted search also found no violation among 39,866 connected bridgeless cubic graphs of order 18. The latter result is exploratory and is not included in the earlier exact rational certificate collection through order 16.

The branch formulation was also tested directly on all 4,597,428 connected simple graphs of minimum degree two and maximum degree three on three through seventeen vertices, as generated by `geng`. All satisfied (4); 618 attained equality. The subset search stops early only after finding a strict lower-bound witness; a reported equality requires exhaustion of all subsets that could improve the score. Completeness up to isomorphism relies on the generator. A separate enumeration checked all 1,225,155 single-edge deletions from the 45,982 connected cubic graphs through order eighteen. These too satisfied the branch inequality. An additional 9,000 degree-preserving local changes to graphs with two degree-two vertices produced no violation.

For three-port fragments, 73,352 vertex deletions from cubic graphs through order sixteen were reduced by componentwise domination and port permutations to 259 representative profiles. An independent standard-library verifier re-enumerates all subsets of each retained fragment and checks its entire boundary table. All 202,020 tested two-fragment joins and 4,900 further heuristic assemblies satisfied the conjectured bound. These computations do not exhaust arbitrary assemblies or larger fragments.

A further enumeration covered all 1,059,165 connected simple graphs on nineteen vertices with exactly one degree-two vertex and all other degrees three. In the notation of Lemma 4.8, each admitted witnesses with $B \geq 9$ and $\max(A, B) \geq 10$. These bounds rule out the doubling construction for those fragments. The search stops once both witnesses are found and does not compute every exact optimum. Separately, 1,225,155 edge deletions from cubic graphs through order eighteen reduce to 27 retained two-attachment tables under componentwise domination and reversal of attachments. The program `verify_boundary_profiles.py` independently enumerates every subset of each retained representative and verifies its complete table and the local inequalities used in Theorem 7.1. The other six retained tables have an infeasible $(2, 2)$ state and are not covered by that theorem. The separate program `verify_matching_criterion.py` checks 451 explicitly constructed good sets in 82 uniform and nonuniform assemblies over K_4 and the Petersen graph, including the averaging identity in the proof.

The separate program `verify_further_reductions.py` checks two failures of stronger proof strategies. The three-connected ten-vertex graph with graph6 code `ICOedO[X?` cannot be covered by two good sets. For the sixteen-vertex graph with graph6 code `O??CA?oBCDT?U?D_CgAA_`, four good sets can cover every vertex at least twice, but every such four-set multiset uses some edge at least three times. Thus imposing an additional edge-use bound of two fails. Their maximum good-set orders are six and ten, respectively, so neither is a counterexample to (1).

Corollary 3.2 leaves the central inequality (4) unproved. The reduction removes neither the unbounded size nor the global compatibility difficulty. No general proof or counterexample is claimed.

15 Disjoint Petersen four-poles

The recent counterexamples to Petersen coloring in [9] provide a separate source of cubic graphs to test. The following argument rules out a broader class of potential counterexamples to (1). No claim of priority is made for this elementary packing argument.

Let W be the Petersen graph with two adjacent vertices deleted. Use vertex set $\{2,3,4,5,6,7,8,9\}$ and edge set

$$\{23, 27, 34, 38, 49, 57, 58, 68, 69, 79\}.$$

Its four degree-two vertices, or ports, are $2, 4, 5, 6$. For every pair of ports there is an induced cycle whose intersection with the ports is exactly that pair:

Selected ports	Vertex set of the induced cycle
$2, 4$	$2, 3, 4, 7, 9$
$2, 5$	$2, 3, 5, 7, 8$
$2, 6$	$2, 3, 6, 7, 8, 9$
$4, 5$	$3, 4, 5, 7, 8, 9$
$4, 6$	$3, 4, 6, 8, 9$
$5, 6$	$5, 6, 7, 8, 9$

The entries are vertex sets, not cyclic orderings. Each has at least five vertices, as can be checked directly from the ten edges of W.

Theorem 15.1. *Suppose a simple cubic graph G contains k pairwise vertex-disjoint induced copies of W. Then $c_{\mathrm{ind}}(G) \geq 5k$. In particular, if $|V(G)| = 8k + r$ and $r \leq 2k$, then G satisfies (1).*

Proof. For each copy create two demand vertices. On the other side of a bipartite graph place one resource vertex for each edge of G leaving at least one copy. Join both demands of a copy to its four leaving edges. These edges are distinct because the copy is induced. For a set of s copies, its $4s$ port incidences use at least $2s$ distinct resources: an edge can meet at most two copies. For any set of demand vertices, let s be the number of represented copies. Its neighbourhood therefore has at least $2s$ resources, whereas it contains at most $2s$ demands. Hall's condition holds.

Choose a matching saturating all demands. In each copy select the cycle in the table corresponding to its two assigned ports. Select no other vertices of G. No leaving edge has both endpoints selected, since that would assign its resource to two copies. Thus each selected vertex has exactly its two neighbours on its local cycle, and the union is induced and 2-regular. It has at least $5k$ vertices. Finally, $5k \geq (8k + r)/2$ when $r \leq 2k$. □

Application and independently checked certificates. The two 52-vertex examples linked in [9], House of Graphs identifiers 57244 and 57278, each contain six disjoint induced copies of W, leaving four other vertices. The theorem gives at least 30 selected vertices, exceeding the required 26. The constructive matching implementation actually returns witnesses of orders 31 and 30, respectively; neither number is asserted to be an optimum. The graph data, six embeddings, and selected vertices are retained in `round19_data/petersen_examples.json`. The standard-library program `verify_round19.py` independently checks simplicity, cubicity, connectivity, every induced embedding, and degree two at every selected vertex. It also enumerates all 256 subsets of W to check the local cycle table. The graph data were retrieved on October 28, 2026 from House of Graphs 57244 and 57278.

Why the obvious smoothing argument stops. Write x_p for the outside neighbour of port p, temporarily assuming these four neighbours are distinct. The natural smoothing removes W and adds x_2x_6 and x_4x_5, when this produces a simple graph. To preserve a chosen outside set, a selected x_p requires port p to have the selection status of its paired outside neighbour. An exhaustive local check of all sixteen outside selection patterns finds an extension with at least five internal vertices in the first fifteen patterns, but no extension when all four outside vertices are selected. The individual witnesses and exact local optima are retained in `round19_verification.json`.

The exceptional case has a short direct explanation. All four ports must then be selected, with exactly one selected internal neighbour. Writing $y_v \in \{0,1\}$ for selection of the four nonports gives

$$y_3 + y_7 = y_3 + y_9 = y_7 + y_8 = y_8 + y_9 = 1.$$

Hence $y_3 = y_8$ and $y_7 = y_9 = 1 - y_3$. If $3, 8$ are selected, vertex 3 has the three selected neighbours $2, 4, 8$. Otherwise $7, 9$ are selected, and vertex 7 has the three selected neighbours $2, 5, 9$. Both cases are impossible. Thus this local smoothing is not an unconditional extension lemma. The packing theorem does not require such an extension, but also does not cover graphs with too few copies of W. The unrestricted conjecture remains unresolved.

16 Three-port certificates and vertex contraction

Let R have odd order q, three designated degree-two ports, and all other vertices of degree three. The fragment need not be connected. A local selected set has degree two at each selected nonport. At each port use state 0 for unselected, 1 for selected with internal selected degree two, and 2 for selected with internal selected degree one. Let $f_R(s)$ be the maximum doubled surplus $2|S| - q$ for boundary state $s \in \{0,1,2\}^3$, with $-\infty$ for an infeasible state. Across an external edge, compatible pairs are exactly $(0,0), (0,1), (1,0), (2,2)$.

Theorem 16.1 (A three-port contraction criterion)**.** *Suppose*

$$f_R(0,0,0) \geq -1,$$

and, for each choice of two ports, there is a state having 2 *at those ports, either* 0 *or* 1 *at the third port, and surplus at least* 1*. Let a simple cubic graph G contain R as an induced fragment, with its three external edges leading to distinct outside vertices. Let G_0 replace R by one vertex adjacent to those three vertices. Then*

$$c_{\mathrm{ind}}(G) \geq c_{\mathrm{ind}}(G_0) + (q-1)/2.$$

Consequently a smallest counterexample to (1) *cannot contain such a fragment with $q > 1$ and distinct outside neighbours.*

Proof. The distinct-neighbour assumption makes G_0 simple and cubic. Take a good set in G_0, and let v be the replacement vertex. If v is unselected, use the $(0,0,0)$ witness in R, adding at least $(q-1)/2$ vertices. Every outside selected vertex sees an unselected port in place of the unselected v.

If v is selected, exactly two of its neighbours are selected. Use the corresponding witness with state 2 at those two ports. The third outside neighbour is unselected, so either state 0 or state 1 at its port is compatible. The local witness has at least $(q+1)/2$ vertices; replacing v therefore gains at least $(q-1)/2$. All selected degrees are two. Taking a maximum good set of G_0 proves the inequality. The order also increases by $q-1$, so the half-bound lifts from the smaller graph. □

The criterion needs only four feasible witnesses, rather than a complete optimum table. It can also be applied successively to substitutions at vertices of any cubic graph already satisfying the conjecture.

Theorem 16.2 (One-port choices in arbitrary assemblies)**.** *Consider disjoint fragments whose ports each receive one external edge, with every external edge joining distinct fragments. Suppose for each fragment either $f_R(0,0,0) \geq 0$, or there is a port p for which*

$$f_R(0,0,0) + f_R(\mathbf{e}_p) \geq 0,$$

where $\mathbf{e}_p$ denotes state 1 at p and 0 at the other ports. Then the resulting cubic graph satisfies (1).

Proof. In fragments with nonnegative all-zero surplus use that witness. Each remaining fragment has two choices: the all-zero witness with surplus a_R, and the one-port witness with surplus b_R, where $a_R + b_R \geq 0$. Two one-port choices conflict only when the two chosen ports are the ends of the same external edge. Each fragment has just one chosen port, so these conflicts form a matching.

For each conflicting pair choose one endpoint uniformly to use its one-port witness and the other to use its all-zero witness. For unpaired fragments choose either witness with probability one half. Every realization is compatible: no external edge has two selected ends. Each variable fragment has expected surplus $(a_R + b_R)/2 \geq 0$, and the fixed fragments have nonnegative surplus. Some realization has nonnegative total surplus. Its local good sets therefore form an induced union of cycles on at least half the vertices of the assembly. □

Larger fragments and certified joins. We computed all vertex-deletion profiles of 42 retained cubic graphs of orders 24 through 32, giving 1,238 indexed fragments. The search program `exploratory/three_pole_large.cpp` enumerates the cycle space of the original graph. A local set with two degree-one ports closes through the deleted vertex; a local set with no such ports is already a union of cycles. These are the only possibilities by the degree-sum formula. Testing induced degrees in the fragment therefore recovers every feasible state, including paths that are not good sets in the original cubic graph.

Of the 1,238 fragments, 1,190 satisfy Theorem 16.1, and 1,230 satisfy Theorem 16.2. The latter 1,230 can be mixed arbitrarily, with arbitrary port orientations, in assemblies of any size subject to that theorem's external-edge condition. The two counts describe overlapping classes, not disjoint classifications. Seven of the eight exceptions to the one-port criterion satisfy the contraction criterion. None of these sufficient conditions is asserted to be necessary.

For pairwise joins, canonically permuting ports leaves 594 distinct tables. Componentwise domination reduces these to 45 retained lower tables. Every ordered pair of retained tables was checked with all six port matchings: 12,150 checks in total. In each case compatible local witnesses have total doubled surplus at least zero; the minimum certified surplus is zero. Domination and port relabelling therefore certify the half-bound for every pairwise join from the full 1,238-fragment

library. These joined graphs have orders 46 through 62. This does not cover arbitrary assemblies of the eight exceptional fragments.

The independent standard-library verifier `verify_round21.py` checks all 13,927 retained feasible-state witnesses directly, the two sufficient criteria, the domination certificates, and all pairwise compatibility checks. These are positive certificates: their validity does not require the table entries to be optimal or the other states to be infeasible. The verifier does not independently establish those upper-bound or infeasibility claims. Data and witnesses are retained in `round21_data`, with verification results in `round21_verification.json`. The contraction criterion still leaves graphs without these fragments, and the original conjecture remains unresolved.

17 The remaining disconnected fragment and mixed assemblies

The unique indexed fragment failing both sufficient criteria of Section 16 is obtained by deleting vertex 27 of its retained 28-vertex source. It is disconnected: its components have orders 14 and 13, with two and one ports respectively. The following exact calculation resolves arbitrary assemblies of copies of this fragment, although mixed assemblies remain a separate question.

Recall the two-port fragment $R = F - e$ from Lemma 6.1, with ports u, v. Form R^+ by adding vertices a, b and edges ua, ab, bv, making a, b the new ports. Then R^+ has fourteen vertices. Its only feasible boundary states are

state	00	11	22
maximum selected vertices	6	8	8
doubled surplus	-2	2	2.

Indeed, in state 00 use a maximum closed state in R, of order six. In state 11, both u, v must be selected with internal degree one in R, giving its state 22 and six vertices, plus a, b. In state 22, the edge ab supplies the internal neighbour of each new port, so u, v must be unselected; use state 00 of R, again with six vertices. A state 01 or 10 is impossible because a saturated selected new port needs both its neighbours, including the other new port. States with exactly one 2 are excluded by parity. These arguments and Lemma 6.1 establish both upper bounds and attainments.

The other component is the thirteen-vertex cap P of Section 2, with root-out and root-in doubled surpluses -1 and 1. Its root cannot have local state 2, again by parity. Explicit isomorphisms from R^+ and P to the two components of the exceptional fragment are retained and independently checked.

Theorem 17.1 (Exact cap-and-chain formula)**.** *Join disjoint copies of R^+ and P by pairing all their ports, one new edge per port, to obtain a simple cubic graph G. Let r_{cyc} be the number of R^+ copies belonging to components of the fragment-incidence multigraph that contain no P copy. Then*

$$c_{\mathrm{ind}}(G) = \frac{|V(G)|}{2} + r_{\mathrm{cyc}}.$$

In particular every such assembly satisfies (1), including every assembly made solely from copies of the exceptional disconnected three-port fragment above.

Proof. In the incidence multigraph, R^+ copies have degree two and P copies have degree one. Thus its components are cycles of R^+ copies or paths of R^+ copies terminated by two P copies. Parallel incidence edges are allowed; an incidence loop at R^+ would duplicate its existing edge ab and is excluded by simplicity.

For a cyclic component with k copies of R^+, use state 22 in every copy. This gives doubled surplus $2k$, which is also an upper bound since each local surplus is at most two.

Consider a path with $k \geq 1$ copies of R^+. A 22 state would propagate along the path to a cap, which has no state 2. Thus each copy uses state 00 or 11. Put $x_i = 0$ or 1 accordingly. Compatibility gives $x_i + x_{i+1} \leq 1$. The maximum contribution of each end cap is $1 - 2x_i$ for its adjacent end copy. The total maximum doubled surplus for this state sequence is

$$\sum_{i=1}^{k}(4x_i - 2) + 2 - 2x_1 - 2x_k = 2\sum_{i=1}^{k-1}(x_i + x_{i+1} - 1) \leq 0.$$

An alternating sequence attains zero; for $k = 1$ the sum is empty and both choices attain zero. A path with no R^+ copy is just two caps joined at their roots, whose maximum doubled surplus is also zero. Local maximizing witnesses combine because their port states are compatible. Adding the contributions over components proves the formula. □

Checks and the remaining mixed case. The independent verifier `verify_round22.py` enumerates all 24,576 vertex subsets of the two components to recover their exact local tables. It checks the decomposition against the original source graph, the two isomorphisms, and 200 independent transfer calculations for paths through length 100 and cycles of lengths two through 100. The theorem itself applies to unbounded lengths.

The separate bounded search `exploratory/exception_assemblies.py` uses all eight exceptional indexed fragments, allowing changes to fragment choices and external port pairings. It retains 388 mixed search records with four through twenty fragments. Another 56 records use uniform choices on fixed cubic bases. Full expanded graph orders range from 100 to 580. For each assembly, a boundary constraint model maximizes total doubled surplus. Of the 444 solver results, 443 are reported optimal and one is feasible only. Every record supplies a nonnegative-surplus witness, and each witness is independently checked for degree two at every selected vertex in the full simple cubic graph. No solver objective is interpreted as an optimum on a merely feasible or unknown result.

All seven uniform assemblies of the disconnected exceptional fragment attain the exact formula above, including the one run whose solver status is only feasible. Optimality in that case follows from the separately proved formula, not the solver status. The remaining mixed records are positive witnesses; their solver optimality claims are not independently certified. Records may repeat an assembly, and the search is not exhaustive. The exact family formula and these mixed-assembly checks do not prove or disprove the original conjecture.

18 Vertex triangles and symmetric bridge obstructions

A remaining inductive question is whether replacing a cubic vertex by a triangle can be the first operation to violate the half-bound. The exact identity below appeared in the earlier computational research record; here we use it to isolate a necessary obstruction and connect it to the symmetric bridge construction.

For a cubic graph G and vertex v, let $G_v^{\triangle}$ replace v by a triangle, with its three new vertices attached bijectively to the three former neighbours of v. Define

$$\begin{aligned} B_v &= \max\{|S| : S \text{ good in } G,\ v \notin S\}, \\ C_v &= \max\{|S| : S \text{ good in } G,\ S \cap N_G[v] = \varnothing\}, \\ D_v &= \max_{e \ni v} \max\{|S| : S \text{ good in } G - e,\ v \in S\}. \end{aligned}$$

An infeasible maximum is $-\infty$.

Lemma 18.1. *With this notation,*

$$c_{\mathrm{ind}}(G_v^{\triangle}) = \max\{B_v, C_v + 3, D_v + 1\}.$$

Proof. A good set meets the new triangle in zero, two, or three vertices; a single selected triangle vertex cannot have two selected neighbours. The zero-vertex case gives exactly B_v. In the three-vertex case, all outside neighbours of the triangle must be unselected, giving C_v+3. In the two-vertex case, both corresponding outside neighbours must be selected. Contract the selected triangle edge back to v and delete the original edge toward the unselected triangle vertex's outside neighbour. The result is good in $G - e$ and contains v. Conversely, every such good set in $G - e$ lifts using the two triangle vertices corresponding to the retained edges at v, gaining one vertex. The outside neighbour on the deleted edge may be selected, which is why $G - e$ cannot be replaced by G in the definition of D_v. □

Theorem 18.2. *Let $|V(G)| = 2h$ and suppose $c_{\mathrm{ind}}(G) \geq h$. The graph $G_v^{\triangle}$ violates (1) if and only if*

$$B_v = h, \qquad D_v \leq h - 1, \qquad C_v \leq h - 3.$$

In that case $c_{\mathrm{ind}}(G) = h$ and $c_{\mathrm{ind}}(G_v^{\triangle}) = h$; no maximum good set of G contains v. Moreover, for every edge e incident with v, subdividing e once to form a rooted graph Q_e and joining two copies at their roots gives

$$|V(D(Q_e))| = 4h + 2, \qquad c_{\mathrm{ind}}(D(Q_e)) = 2h.$$

Thus every such failed triangle expansion also produces a symmetric bridged counterexample to the original conjecture.

Proof. Every good set of G containing v remains good after deletion of its unique incident edge toward an unselected neighbour. Its size is therefore at most D_v. Consequently $c_{\mathrm{ind}}(G) \leq \max\{B_v, D_v\}$. If the expansion violates the half-bound, its order is $2h+2$ and Lemma 18.1 gives $B_v \leq h$, $D_v \leq h-1$, and $C_v \leq h-3$. The assumed lower bound on G forces $B_v = h$ and $c_{\mathrm{ind}}(G) = h$. The converse and the exact expanded value follow directly from the same identity. The bound on D_v excludes v from every maximum good set.

Fix an incident edge $e = vu$. In $G - e$, a good set omitting v has maximum size $B_v = h$, while one containing v has size at most $D_v \leq h - 1$. Hence $c_{\mathrm{ind}}(G - e) = h$. A good set of G containing both endpoints of e has size at most $h - 1$, since it contains v. After subdivision, the root-out optimum is therefore $B = h$, and the root-in optimum is $A \leq h$. The exact doubling identity of Lemma 4.8 yields $\max\{2B, A + B\} = 2h$, on $4h + 2$ vertices. □

Corollary 18.3. *If $c_{\mathrm{ind}}(G) > |V(G)|/2$, every single vertex-to-triangle expansion satisfies (1). The same is true at a vertex belonging to a good set of size at least $|V(G)|/2$. In particular, if $V(G)$ partitions into two good sets, every single such expansion satisfies the bound.*

Proof. For a strict half-bound witness, keep it if v is absent and lift it with a gain of one if v is present. A half-sized witness containing v also lifts with a gain of one. In a partition into two good sets, every vertex has two neighbours in its own part and one in the other part. Counting crossing edges shows that the parts have equal size. Every vertex thus belongs to a half-sized witness. □

Verification and a failed shortcut. Independent enumeration checks Lemma 18.1 for all 246 vertex expansions of the 27 connected cubic graphs through order ten. Separately, complementary good-set partitions were found and directly verified for 100 previously certified sharp records: five cubic records and 95 symmetric joins of rooted records. Their orders range from 24 to 58; distinct

records need not be nonisomorphic. The verifier explicitly expands every vertex and checks the lifted set, giving 3,932 verified half-bound witnesses. This does not assert that every sharp cubic graph admits such a partition, or that repeated expansions preserve it.

One possible shortcut would assert that no cubic vertex can have all three incident edges increase c_{ind} when deleted. This assertion is false. The fourteen-vertex graph with graph6 code `M???FBOq@KF?D_B_?` has $c_{\mathrm{ind}} = 8$, whereas deleting any of $\{2,7\}$, $\{2,10\}$, and $\{2,11\}$ raises the value to 10. Exhaustive vertex-subset enumeration independently checks these values and retains explicit witnesses. The graph satisfies the original half-bound, since $8 > 14/2$; it refutes only that auxiliary shortcut. It does not settle the more restrictive sharp-graph case.

The certificate generator is `exploratory/triangle_obstruction_certificates.py`, and the independent verifier is `verify_round23.py`. Data are retained in `round23_data`. Theorem 18.2 is a conditional reduction, not a proof that its obstruction is impossible. The original conjecture remains unresolved.

18.1 Direct searches for the triangle obstruction

We next applied the preceding criterion to every cubic parent in the two radius-four edge neighbourhoods of Section 14.7. For each parent, a good set larger than half certifies every single triangle expansion. Otherwise, half-sized good sets whose union covers all parent vertices suffice: expand a vertex using a witness that contains it. The implementation tests previously retained witnesses and, when necessary, uses exhaustive vertex-assignment search to find additional ones. If these criteria fail, it directly checks the expanded graph; no such fallback was needed in the completed runs.

All single-vertex expansions in both regions satisfy the half-bound. The 24-vertex region has 1,974,349 reconnection records and hence 47,384,376 indexed expansions. The 26-vertex region has 2,897,883 records and 75,344,958 indexed expansions. In total this is 122,729,334 indexed expansions. These counts include duplicate constructions and must not be interpreted as numbers of distinct labelled or nonisomorphic graphs.

The calculation retained 5,656 additional strict parent witnesses and 76 additional half-bound witnesses. The independent standard-library verifier checks all 5,732 directly and constructs every triangle lift justified by each witness: all parent vertices for a strict witness, and its selected vertices for a half-bound witness. This gives 142,886 explicit verified lifts. The large reconnection regions are not independently regenerated; completeness relies on the C++ enumeration, whose generation scheme was separately checked against complete small labelled graph lists in Section 14.7. The implementation is `exploratory/triangle_edge_ball.cpp`.

A broader constraint-generation search directly targets a cubic counterexample with one prescribed triangle. Its parents have orders 24 or 26, fixed neighbours at the expanded vertex, and at most six removed edges relative to a relabelled sharp seed. Each discovered half-sized good set in the expanded graph becomes a forbidden-set constraint for subsequent candidates. The two runs produced eight and six candidates, accumulating 1,821 and 1,907 constraints, then stopped at wall-time limits. Independent cycle-space enumeration checks every candidate's reported maximum and number of half-bound witnesses; it also verifies that each parent satisfies the original bound. The timeouts exclude no additional region. The source is `exploratory/triangle_counterexample_synthesis.py`.

Finally, all 21 edge deletions of the fourteen-vertex critical-star example were joined to the 27 retained two-port profiles, in both orientations. All 1,134 boundary-table checks have doubled surplus at least two. Thus this particular attempt to turn the auxiliary critical-star example into a sharp graph or an original counterexample also fails. It is a finite fragment calculation, not a theorem about arbitrary partners.

The original conjecture remains unresolved; see `round24_verification.json`.

19 Repair barriers with only a total edge bound

The pseudoforest restriction is not responsible for all the barriers in Section 13. Allow arbitrary odd spanning factors as intermediate states, subject only to $|E(M)| \le n$. This is the relevant edge bound because an isolated-edge-free odd factor with at most n edges gives a good set of size at least $n/2$. Define $h_G^{\mathrm{edge}}(M)$ as the minimum of $\max_i j(M_i) - j(M)$ over such simple-cycle exchange sequences ending at a factor with smaller j. There is no restriction on the cycle ranks of their components.

Theorem 19.1. *For the graphs B_k and initial factors M_k of Section 13,*

$$\left\lceil \frac{k-1}{4} \right\rceil \le h_{B_k}^{\mathrm{edge}}(M_k) \le k-1 \qquad (k \ge 3).$$

In particular, allowing components with arbitrarily many cycles does not permit repairs with any fixed bound on the temporary increase in isolated edges. Nevertheless, $c_{\mathrm{ind}}(B_k) = |V(B_k)|/2$.

Proof. Consider any restriction N of an odd factor to a cap before either displayed k-cycle has been exchanged. Its root degree is zero or two, and its other twelve degrees are odd. Retain the notation $w(N) = j_P(N) + \mathbf{1}_{\{\deg_N(r)=0\}}$, but impose no pseudoforest condition. We claim

$$|E(N)| \ge 13 - 4w(N). \tag{16}$$

If $w(N) = 0$, the root has degree two and there is no isolated edge. The nonroot degree-one vertices L induce a good set in P: edges outside N give two neighbours in L to each such vertex, and an edge of N between two vertices of L would be isolated. The cap bound gives $|L| \le 6$, and the degree sum gives $|E(N)| = 19 - |L| \ge 13$. For $w(N) \ge 1$, write d for the number of nonroot degree-three vertices and $\beta \ge 0$ for the total cycle rank of N. If the root has degree two, then $m = |E(N)| = 7 + d$, so N has $13 - m + \beta = 6 - d + \beta$ components. Apart from the root component, every component that is not an isolated edge contains a degree-three vertex. Hence $j_P(N) \ge 5 - 2d + \beta$, giving $2m + w(N) \ge 19$. If the root is isolated, then $m = 6 + d$ and the nonroot subgraph has $6 - d + \beta$ components. At most d of these are not isolated edges. Thus $j_P(N) \ge 6 - 2d + \beta$ and again $2m + w(N) \ge 19$. Consequently $m \ge (19 - w(N))/2 \ge 13 - 4w(N)$ for $w(N) \ge 1$.

Every bridge is present in every odd factor: summing odd degrees on either side of a bridge in a cubic graph forces its inclusion. Moreover, a simple cycle cannot cross a bridge. Until the first displayed-cycle exchange, both displayed cycles therefore remain absent and the central bridge remains an isolated edge. Writing N_i for the $2k-2$ cap restrictions and $W = \sum_i w(N_i)$, we have

$$j = 1 + W, \qquad |E(M)| = (2k-1) + \sum_i |E(N_i)| \ge n_k - 1 - 4W.$$

Some displayed cycle must eventually be exchanged to eliminate the central isolated edge. Its first exchange adds exactly k edges. The edge bound immediately after it requires $n_k - 1 - 4W + k \le n_k$, whence $4W \ge k - 1$. This proves the lower bound on the peak excess.

The explicit k-exchange sequence in Section 13 has edge counts $n_k - 1, n_k - 2, \dots, n_k - k, n_k$ and defect counts $1, 2, \dots, k, 0$. It obeys the total edge bound, proving the upper bound. The equality for $c_{\mathrm{ind}}(B_k)$ was proved there. □

An independent finite check confirms the local bound. Exhausting its 2^{19} edge subsets leaves precisely 128 states with the required degree parities. Their minimum edge counts at each cost are

w	0	1	2	3	4	5	7
$\min \lvert E(N)\rvert$	13	9	9	8	8	7	6

The theorem is independent of this enumeration. The table also shows that the coefficient four in the local bound cannot be reduced: a state with $w = 1$ and nine edges attains equality. The standard-library program `verify_round25.py` checks the table directly, the root-avoiding cap maximum, and all edges and degrees in the explicit repairs for $3 \le k \le 40$.

This rules out a bounded-defect cycle-exchange strategy even after removing the componentwise pseudoforest constraint. It does not exclude strategies allowing unbounded temporary defects, and it does not settle the existence of an isolated-edge-free odd factor with at most n edges in an arbitrary cubic graph. The original conjecture remains unresolved.

20 Arbitrary assemblies of four-port fragments

The following extension of the Hall argument in Section 15 rules out another direct construction of counterexamples. A four-port fragment R has four distinct degree-two vertices, called ports; every other vertex has degree three. Its order q is even. The fragment need not be connected. Call a set *closed* if it induces a 2-regular subgraph of R, so that every selected port already has two selected neighbours within R.

Theorem 20.1 (A sufficient condition for arbitrary four-port assemblies)**.** *Suppose each member R_i of a collection of disjoint four-port fragments, of orders q_i, satisfies at least one of these conditions:*

1. *There is a closed set of size at least $q_i/2$ avoiding all ports.*
2. *For each of the six pairs of ports, there is a closed set of size at least $q_i/2$ whose selected ports are exactly that pair.*

Pair all ports by edges between distinct fragments, obtaining a simple cubic graph G. Then $c_{\mathrm{ind}}(G) \ge |V(G)|/2$, for any number of fragments and any such pairing.

Proof. Use the first condition wherever available, and call the remaining fragments active. Make a bipartite graph with two demand vertices for each active fragment on the left, and one resource vertex for each joining edge on the right. Both demands of a fragment are adjacent to its four joining edges. For any set of demands, let s be the number of represented fragments. Their $4s$ port incidences belong to at least $2s$ distinct joining edges, because each edge has at most two ends in these fragments. The number of demands is at most $2s$. Hall's condition therefore holds, and a matching assigns two distinct joining edges to every active fragment, with no edge assigned twice.

In an active fragment select the closed set for the two ports incident with its assigned edges. In every other fragment select the port-avoiding closed set. No joining edge has both endpoints selected: that would require it to be assigned to both endpoint fragments. Thus the union of these closed sets is good in G and has size at least $\sum_i q_i/2 = |V(G)|/2$. □

We tested fragments obtained by deleting adjacent vertices u, v with no common neighbour from 56 retained sharp cubic examples: the Franklin graph, the five examples of orders $24, 26, 36, 38, 42$, and the symmetric doubles of the fifty retained thirteen-vertex caps from Section 18. These are labelled records, not a claimed nonisomorphic census. There are 2,214 eligible edge-deletion records. Of these, 1,643 satisfy the first condition and all remaining 571 satisfy the second. Consequently *every* assembly made solely from this library satisfies the conjecture, allowing arbitrary repetition, orientation, and number of pieces. This conclusion goes beyond a finite search over pairwise joins, but does not cover arbitrary four-port fragments.

For completeness, the exact profile generator enumerates Eulerian edge sets of the original cubic graph, removes u, v from their supports, and checks induced degrees in the fragment. Each port is

unselected (state 0), selected with internal degree two (state 1), or selected with internal degree one (state 2). The resulting 81-entry table records the maximum doubled surplus $2|S| - q$ at each state. This enumeration is exhaustive: a feasible partial cycle system has zero, two, or four state-2 ports. With two, close its ends through u, v, or the path uv, according to their locations. With four, join the two ports formerly adjacent to u through u and the other two through v. With zero, no added edge is needed. In each case the resulting edge set is Eulerian in the original graph. It need not be induced there; inducedness is checked only after deletion.

The independent verifier checks all 44,154 feasible-state witnesses directly and verifies that every eligible deletion of each saved seed is included. Only positive witnesses are used for Theorem 20.1; large-profile maxima and infeasibility claims are not needed for this certificate. As a separate generator check, all 252 eligible profiles from connected cubic graphs through order ten agree with exhaustive vertex-subset enumeration. The verifier also constructs 41 complete assemblies and checks their good sets directly, including an assembly using every retained piece with 54,996 vertices. These checks test the construction; its general validity follows from the Hall proof. The sources are `exploratory/four_pole_large.cpp` and `verify_round26.py`; results are saved in `round26_verification.json`. The original conjecture remains unresolved outside the proved classes.

21 A four-port contraction criterion

The preceding assembly theorem can be complemented by a local reduction that applies even when the rest of the graph is arbitrary. Use the boundary states $0, 1, 2$ of Section 20, and write f_s for the maximum doubled surplus $2|S| - q$ in state s of a four-port fragment of order q, with value $-\infty$ for an infeasible state. The following conditions can equally be certified by individual witnesses; exact maxima are not needed.

Theorem 21.1 (Replacing a fragment by two adjacent vertices)**.** *Let R be an induced four-port fragment in a simple cubic graph G, with ports ordered p_1, p_2, p_3, p_4. Suppose*

1. *$f_{0000} \geq -2$;*
2. *$f_{2200} \geq 0$ and $f_{0022} \geq 0$;*
3. *for every $i \in \{1, 2\}$ and $j \in \{3, 4\}$, some state with entries $s_i = s_j = 2$ and the other two entries in $\{0, 1\}$ has $f_s \geq 2$.*

Delete R, add adjacent vertices u, v, and reattach the former external edges at p_1, p_2 to u and those at p_3, p_4 to v. Assume the resulting graph G_0 is simple and cubic. Then

$$c_{\mathrm{ind}}(G) \geq c_{\mathrm{ind}}(G_0) + \frac{q-2}{2}.$$

Consequently a minimum-order counterexample to the original conjecture contains no such fragment of order $q > 2$ for which the specified replacement gives a simple cubic graph.

Proof. Fix a good set S_0 in G_0. If neither u nor v belongs to S_0, replace them by a state-0000 set in R. It has at least $(q-2)/2$ vertices and has no selected port, so it has no edge to the retained selected vertices outside R.

If only u is selected, both its external neighbours must be selected. Use a state-2200 set, with at least $q/2$ vertices. Its two selected ports have internal degree one and receive their second selected neighbour outside R. The other ports are unselected. All selected exterior vertices retain exactly

the same number of selected neighbours. Replacing the one selected vertex u therefore gains at least $(q-2)/2$ vertices. The case where only v is selected uses state 0022.

If both u, v are selected, each has exactly one selected external neighbour. Let their corresponding ports be p_i, p_j, one in each pair. Use the state from the third condition. Its open ports attach to the selected exterior neighbours. Either of the other ports may be selected with internal degree two, since its exterior neighbour is unselected. Again all selected degrees are exactly two. This replaces two selected vertices by at least $(q+2)/2$, giving the same gain. Maximizing over S_0 proves the inequality. Since $|V(G)| = |V(G_0)| + q - 2$, it transfers the conjectured half-bound from G_0 to G and proves the minimal-counterexample assertion. □

With the port pairing inherited from the deleted adjacent vertices, 1,945 of the 2,214 fragments in Section 20 meet this criterion. The other 269 are not covered by this contraction test. These counts do not assert that any cubic graph contains a qualifying fragment, nor that its contraction is always simple. Directly replacing an edge and its endpoints in $K_{3,3}$ by each certified fragment checks 19,450 lifts, covering all good sets of that six-vertex exterior model. The general statement follows from the preceding case proof.

We also broadened the direct counterexample search to all eligible adjacent-vertex deletions from the 621 connected simple cubic graphs of orders at most fourteen. This produces 9,492 indexed four-port fragments, whose orders range from four to twelve. Their tables have 7,144 distinct labelled value vectors and 2,054 vectors up to port permutation. Retaining 1,239 vectors suffices under componentwise domination and port permutation. Every discarded vector dominates some retained vector in a suitable port order. Thus a nonnegative joining witness for the retained vectors transfers to the discarded ones: replacing either profile by a componentwise larger profile cannot decrease the best joined surplus.

All $24\binom{1240}{2} = 18,436,320$ pairings of retained profiles with repetition have a nonnegative joining witness. Hence every two-fragment join from this entire library, in every port orientation, satisfies the original half-bound. The resulting orders are at most 24. These are indexed constructions, with possible repeated graphs; they are not a census of cubic graphs on those orders. This finite result does not imply that arbitrary larger assemblies from this broader library satisfy the bound.

The verifier `verify_round27.py` checks all 210,400 recorded local witnesses by induced degrees, every discarded-profile domination, and all retained pairings. Its pair checker uses compatible-state bitsets grouped by score, independently of the exploratory C++ enumeration of the 256 compatible state pairs for each orientation. Large-profile upper bounds are not needed for these positive checks. Data and results are in `round27_data/` and `round27_verification.json`. The original conjecture remains unresolved.

22 Assemblies using open ports and a two-factor

The closed-port condition of Section 20 is not the only way to assemble fragments. Keep the same states and doubled-surplus profiles. For a pair I of ports, let $s(I)$ be the state with entries two at I and zero at the other ports. Write e_j for the state vector with entry one only at port j.

Theorem 22.1 (Open-port assembly criterion)**.** *Suppose every fragment R in a collection of four-port fragments satisfies the following condition for every pair I of its ports:*

$$f_{s(I)} \geq 0 \quad \text{or} \quad f_{s(I)+e_j} \geq 0 \text{ for both } j \notin I.$$

Any simple cubic graph obtained by pairing all ports between distinct fragments satisfies $c_{\mathrm{ind}}(G) \geq |V(G)|/2$.

Proof. Contract each fragment to a vertex, retaining the joining edges. The resulting loopless multigraph Q is 4-regular. It has a 2-factor F: orient an Euler tour in each component, giving indegree and outdegree two, and take a perfect matching in the associated bipartite graph with one outgoing and one incoming copy of each vertex. This bipartite graph is 2-regular, so Hall's theorem provides the matching. Its directed edges give the required 2-factor.

In each fragment the two ports incident with F will have state two. If state $s(I)$ has nonnegative surplus, use it. Call the remaining fragments active. Each active fragment can instead use either of its two non-F ports in state one, leaving the other in state zero. Assign one incident non-F edge to each active fragment, without assigning any edge twice. Hall's theorem again applies: any s active fragments have $2s$ incidences with non-F edges, belonging to at least s distinct edges.

Use the state-one port on the assigned edge. A joining edge in F has state two at both ends; it completes each selected endpoint's internal degree one to degree two. A joining edge outside F has at most one selected endpoint, since it is assigned at most once. It therefore adds no selected neighbour to a state-one port, whose internal degree is already two. The union of the local sets is good, and every fragment contributes at least half its vertices. □

Of the 9,492 fragments in Section 21, 5,552 satisfy this condition. Thus arbitrary assemblies using only those fragments satisfy the half-bound. The independent verifier checks their local witnesses and constructs 21 whole-graph examples using Euler orientations and two bipartite matchings. One uses every qualifying fragment and has 65,380 vertices. Its selected set is checked by actual induced degrees. This is a second sufficient assembly criterion; arbitrary mixtures with fragments satisfying only Theorem 20.1 are not claimed to be covered.

22.1 Direct cyclic-assembly searches

We also searched assemblies whose incidence multigraph is a cycle with two parallel edges between consecutive fragments. Each fragment has two left and two right ports. If $a, b \in \{0,1,2\}^2$, define the 9-state transfer matrix

$$M_R(a,b) = \max_{t \sim a} f_R(t,b),$$

where compatibility is coordinatewise and permits exactly the pairs $(0,0),(0,1),(1,0),(2,2)$. With max-plus matrix multiplication, the optimum doubled surplus of a fixed cyclic assembly is

$$\max_a (M_{R_1} M_{R_2} \cdots M_{R_k})(a,a).$$

This is exact because a compatible state choice determines disjoint local solutions, and the boundary conditions make their union good.

The retained 1,239 profiles and all 24 port permutations give 29,736 indexed choices per position. Coordinate descent scans all choices at one position while keeping the others fixed. The recorded search uses $k \in \{3,4,5,6,8,12,20,40\}$, three restarts per length and five sweeps per restart: 43,711,920 coordinate evaluations and 1,470 retained assembly records. This is a heuristic search over words of fragments, not exhaustive enumeration of all such words. Independent integer dynamic programming reconstructs and verifies every recorded value and a whole-graph good set. Their orders range from 12 to 470, and none violates the half-bound.

All eight zero-surplus words are Franklin relabellings, verified by explicit isomorphisms. CP-SAT separately excludes larger good sets; complementary good partitions are checked by induced degrees. Data are in `round28_data/`; independent checks are implemented in `verify_round28.py` and `verify_round28_assembly.py`. The original conjecture remains unresolved.

22.2 Mixing the two assembly criteria

The two criteria can be combined when the joining graph supplies a suitable collection of cycles. This is a sufficient condition, not an assertion that all mixtures are covered.

Theorem 22.2 (Mixed assembly criterion)**.** *Let Q be the loopless 4-regular multigraph of joining edges of a four-port assembly. Suppose Q has a subgraph F in which each vertex has degree zero or two. At every degree-zero vertex, require the closed-port condition of Theorem 20.1. At every degree-two vertex, require the open-port condition of Theorem 22.1. Then the assembled simple cubic graph G satisfies $c_{\mathrm{ind}}(G) \geq |V(G)|/2$.*

Proof. For a degree-zero fragment, choose its port-avoiding solution if available, creating no demand. Otherwise it can select any pair of ports and demands two distinct joining edges. For a degree-two fragment, give state two to the ports on F. It either has a nonnegative-surplus solution avoiding both remaining ports, or can use either remaining port in state one and demands one non-F edge.

All demands are assigned to edges of $Q - F$, with no edge assigned twice. Indeed, at a degree-zero vertex $Q - F$ has degree four and there are at most two demands; at a degree-two vertex it has degree two and there is at most one demand. For any set of demand slots, let W be the represented vertices. The number of slots is at most $\frac{1}{2}\sum_{v \in W} d_{Q-F}(v)$, whereas the number of incident resource edges is at least this quantity. Hall's theorem supplies the assignment. Use exactly the assigned ports in state one.

Edges of F have state two at both ends and complete the selected degrees to two. Every other joining edge has at most one endpoint in state one and neither endpoint in state two. Thus all selected degrees are two, and the local nonnegative surpluses sum to a nonnegative surplus for G. □

Taking F empty recovers the closed-port theorem; taking it spanning recovers the open-port theorem. In particular, for a K_5 joining graph, any mixture with three or four designated open-type fragments is covered: take a cycle through exactly those vertices. With only one or two designated open-type fragments, this choice of F is impossible. This failure of the sufficient condition is not a counterexample to the original conjecture. The verifier constructs and degree-checks 1,000 mixed examples of orders 44 through 60, using fragments that satisfy only their designated criterion.

A separate counterexample-guided search fixes the K_5 joining graph and allows ten closed-type and ten open-type profiles, each in all 24 port orders. It fixes one block to each type, to require a genuine mixture. Each nonnegative whole-graph witness supplies an exclusion constraint on the profile choices. After 120.7 seconds the run reached its time limit with 41 such witnesses and 78 solver queries. The witnesses have orders 36 through 60 and doubled surplus at least twelve. The time limit does not prove the remaining restricted domain infeasible. All saved witnesses and their boundary compatibilities are checked directly by `verify_round29.py`.

Finally, a focused exhaustive check uses only the four-cycle fragment and the eight-vertex fragment obtained from graph6 `I?BeeOwM?` by deleting vertices 2 and 5 (zero-based labels). The latter has $f_{0000} = -8$ and all six closed-pair values at least zero. Each piece has three distinct boundary profiles under its 24 port permutations. For one or two four-cycle pieces and all remaining pieces of the second type on K_5, there are $2 \cdot 3^5 = 486$ indexed profile assignments. Every one has a directly checked good set with doubled surplus at least ten. The graphs have orders 32 or 36. This exhausts these two selected fragment types on K_5, not the wider mixed library. The source is `exploratory/mixed_k5_exhaust.py`; the original conjecture remains unresolved.

23 Two-edge extensions and the order-22 census

Let $e = uv$ and $f = wz$ be disjoint edges of a simple cubic graph G. Let $G^+_{e,f}$ be obtained by subdividing e, f with new vertices x, y, respectively, and adding xy. This increases the order by two and preserves simplicity and cubicity. Unlike vertex-triangle expansion, the operation need not introduce a triangle.

Define $B = c_{\mathrm{ind}}(G - e - f)$. Let T_e be the largest good-set size in $G - f$ subject to containing both u, v, and define T_f symmetrically in $G - e$. Finally, let P be the largest size of a vertex set S in $G - e - f$ such that exactly one endpoint of each of e, f belongs to S, these two selected endpoints have induced degree one, and every other vertex of S has induced degree two. Thus S induces a path between the two selected endpoints, together with possibly some cycles. Infeasible constrained maxima have value $-\infty$.

Proposition 23.1 (Exact two-edge extension identity)**.**

$$c_{\mathrm{ind}}(G^+_{e,f}) = \max\{B, T_e + 1, T_f + 1, P + 2\}.$$

Proof. Separate good sets according to their intersection with $\{x, y\}$. If neither is selected, their restriction to the original vertices is exactly a good set of $G - e - f$. If only x is selected, both u, v must be selected; restoring the edge uv in place of the selected path uxv gives a good set of $G - f$. This correspondence is reversible and changes the size by one. The case where only y is selected is symmetric. If both are selected, each has exactly one selected old neighbour. Removing x, y leaves precisely a set counted by P. Conversely, adding x, y and their joining edge closes its path into an induced cycle, leaving its other cycle components unchanged. This last correspondence changes the size by two. □

For $|V(G)| = 2h$, this construction would violate the original conjecture exactly when

$$B \le h, \qquad T_e, T_f \le h - 1, \qquad P \le h - 2.$$

These are exact conditions for this construction, not their exclusion. We tested every disjoint edge pair in 101 retained sharp parent records: the Franklin graph and the 100 records used in Section 18. All 182,679 extensions have directly verified half-bound witnesses. Their orders range from 14 to 60; repeated constructions are possible. Independently, the displayed identity agrees with exhaustive vertex-subset enumeration in all 1,674 eligible edge-pair cases from connected cubic graphs through order ten. Data are in `round30_data/`; the verifier is `verify_round30_extensions.py`.

23.1 Complete connected census at order twenty-two

The unrestricted computation at order 22 uses nauty 2.9.3 with command `geng -c -d3 -D3 -X0 -x3200 22 r/64`, for all residues $r = 0, \ldots, 63$. The 64 disjoint shards contain 7,319,447 connected simple cubic graphs up to isomorphism, agreeing with the published House of Graphs count [10]. For every graph we retain a vertex mask whose selected vertices induce a 2-regular graph of order at least twelve. The masks and their graph6 inputs are stored in compressed per-shard certificate files.

A separate C++ checker imports no search routines. It decodes each graph, checks all degrees and connectivity, and directly counts the selected neighbours of every selected vertex. Every certificate passes. The generator's count, the search count and the independent checker count agree for every shard. Enumeration completeness relies on `geng`'s isomorph-free generation and documented residue splitting; the degree check independently establishes the positive bound for every retained graph. This is a computational census, not a general proof.

The bound twelve is attained. The graph with graph6 string

```
U??????????[?s?q?Co@o?DO?K_?Co?F??@W??M??
```

has a good set $\{0, 2, 3, 6, 7, 9, 11, 12, 14, 16, 18, 20\}$, using zero-based labels. Independent enumeration of all $2^{22} = 4,194,304$ vertex subsets gives

$\lvert S\rvert$	0	4	6	8	10	12
# good sets	1	6	18	24	66	84

and no larger good set. Thus the minimum of $c_{\mathrm{ind}}(G)$ over connected simple cubic graphs of order 22 is exactly twelve. Together with the previous smaller-order checks, any counterexample to the original conjecture must have at least 24 vertices: a disconnected counterexample would have a connected component that is itself a counterexample. The programs `verify_round30_census.py` and `verify_round30_minimum.py` record the checks and certificate hashes. No proof or counterexample for arbitrary orders has been found.

24 A two-edge extension can lose arbitrarily many vertices

The exact identity in Section 23 does not imply an additive induction step. In fact the optimum can decrease without any uniform bound under a single two-edge extension. We give an explicit infinite family and explain why it still satisfies the original conjecture.

24.1 An attachment that preserves the unweighted optimum

Let Q be the triangular prism with triangles abc and $a'b'c'$ and matching edges aa', bb', cc', with the edge ab subdivided by a root z. The root has degree two and the other six vertices have degree three.

Lemma 24.1. *The largest good sets in Q containing and avoiding z both have size five.*

Proof. The cycles $azbb'a'a$ and $acbb'a'a$ are induced five-cycles, containing and avoiding z, respectively. A six-vertex good set would contain at least five of the six degree-three vertices. Each of those vertices would need a neighbour outside the set. The unique outside vertex has degree at most three, a contradiction. The full seven-vertex set is not good either. Thus five is an upper bound as well as an attained value. □

For a subcubic graph H, attach a separate copy of Q at each missing incidence, joining its root to the deficient vertex. Write $\mathcal{N}(H)$ for the resulting cubic graph and $D = 3|V(H)| - 2|E(H)|$.

Proposition 24.2. *For every subcubic graph H,*

$$|V(\mathcal{N}(H))| = |V(H)| + 7D, \qquad c_{\mathrm{ind}}(\mathcal{N}(H)) = 5D + c_{\mathrm{ind}}(H).$$

Proof. Each new joining edge is a bridge, so no good set uses it. The restriction of a good set to H is good, and each attached copy contributes at most five vertices by Lemma 24.1. Conversely, any good set in H extends by choosing a root-avoiding five-cycle in every copy. This is valid whether the corresponding vertex of H is selected or not. Both bounds are attained together. □

This differs from the branch-counting completion in Section 3: it preserves the unweighted optimum up to an additive constant. It is useful for transferring sensitivity examples from subcubic cores to cubic graphs, but its surplus does not preserve the conjectured half bound.

24.2 An exact unbounded loss

For $k \geq 2$, form G_k by taking a cycle $v_0v_1 \cdots v_{2k-1}v_0$ and attaching a copy of Q at each vertex. Let G_k^+ be its two-edge extension at the disjoint edges v_0v_1 and v_kv_{k+1}, with new vertices x, y and edge xy.

Theorem 24.3. *These graphs are connected, simple and cubic, and satisfy*

$$|V(G_k)| = 16k, \qquad c_{\mathrm{ind}}(G_k) = 12k,$$
$$|V(G_k^+)| = 16k + 2, \qquad c_{\mathrm{ind}}(G_k^+) = 11k + 2.$$

Consequently $c_{\mathrm{ind}}(G_k) - c_{\mathrm{ind}}(G_k^+) = k - 2$*, which is unbounded.*

Proof. There are $2k$ attachments, each contributing at most five vertices, and a root-avoiding optimum is always available in each. Their joining edges remain bridges after the extension. Thus in both graphs the optimum is $10k$ plus the optimum of the remaining core.

The original core is an induced cycle on $2k$ vertices. The new core is a theta graph with three internally disjoint x–y paths: the edge xy and two paths each of length $k+1$. Every cycle in a theta graph uses two of its three paths. The cycle using both long paths is not induced, because xy is a chord. Each cycle using xy and one long path is induced and has $k+2$ vertices. No two cycles are vertex-disjoint, since both would contain x, y. Hence the largest induced 2-regular subgraph of the core has exactly $k+2$ vertices. Adding the attachment contributions gives the two formulas. □

More generally, if a cycle has length L and the two subdivided edges have cyclic separation d, where $2 \leq d \leq L-2$, the extended core has optimum $\max(d, L-d) + 2$. This gives a loss of $\min(d, L-d) - 2$ after attaching the same gadgets.

The theorem refutes both the proposed inequality $c_{\mathrm{ind}}(G_{e,f}^+) \geq c_{\mathrm{ind}}(G) + 1$ and any proposed universal lower bound $c_{\mathrm{ind}}(G_{e,f}^+) \geq c_{\mathrm{ind}}(G) - C$ with a fixed constant C. It does not refute the original conjecture: the exact surplus in the extended family is

$$c_{\mathrm{ind}}(G_k^+) - |V(G_k^+)|/2 = 3k + 1 > 0.$$

An induction based on either of these additive inequalities is therefore invalid. The boundary quantities in Section 23 retain information that those inequalities discard; no sufficient general induction step has been established here.

24.3 Independent checks and smaller examples

Exhaustive vertex-subset checks give two smaller failures of additive monotonicity. Edge endpoints use zero-based graph6 labels.

Parent graph6	Extended edges	Orders	Before	After
`K?ABCN_U?wF?`	$(0,7), (3,10)$	$12 \to 14$	9	8
`O???C@_sAaSGIGB_@o?k?`	$(0,7), (8,12)$	$16 \to 18$	12	10

Here "before" and "after" are exact maxima, not just witness sizes. Both extended graphs satisfy the original half bound. The program `verify_round31.py` independently enumerates all subsets of these four graphs and of the seven-vertex attachment. It also checks the theta-core formula exhaustively through core order $L = 18$ and constructs whole-graph witnesses for $2 \leq k \leq 100$. The infinite-family upper bound follows from the preceding proof, not from extrapolating these computations. The original conjecture remains unresolved.

25 Edge-boundary certificates for bridge joins

A decrease in the full optimum under an extension does not by itself threaten the half bound. We now test a sufficient boundary condition that certifies that bound directly for rooted subdivisions and their bridge joins. The condition is not proved for arbitrary cubic graphs.

Proposition 25.1. *Let G be a connected simple cubic graph of order $2h$, and let S be a good set of size $c \geq h+1$. Suppose that for every edge $e \in E(G[S])$ there is a good set T_e in $G-e$ with*

$$|T_e| \geq 2h - c.$$

For every edge e of G, let Q_e be obtained by subdividing e once and taking its new vertex as the root. Then $c_{\mathrm{ind}}(D(Q_e)) \geq 2h+1$ for every e. Moreover, joining the roots of any two fragments certified in this way, possibly from different parent graphs of different orders, satisfies the original half bound.

Proof. If $e \notin E(G[S])$, subdividing it leaves S good with the root unselected. Hence the root-out optimum B_e is at least $c \geq h+1$, and $2B_e \geq 2h+2 > 2h+1$.

If $e \in E(G[S])$, subdividing its selected edge gives a root-in set of size $c+1$. The set T_e gives a root-out set of size at least $2h-c$. Thus

$$A_e + B_e \geq (c+1) + (2h-c) = 2h+1.$$

In either case the doubling formula in Lemma 4.8 certifies the half bound for $D(Q_e)$. The final assertion follows from bridge symmetrization, Theorem 4.9: if each individual symmetric join meets its half bound, so does any mixed join of the two rooted fragments. □

The strict inequality $c > h$ is essential to this particular certificate: when $c = h$ and $e \notin E(G[S])$, the available root-out bound gives only $2B_e \geq 2h$, one below the required order $2h+1$. The proposition does not assert that its boundary hypothesis always holds, even when a strict good set is known.

25.1 Exhaustive verification for order-22 parents

We applied Proposition 25.1 to the saved good set in every one of the 7,319,447 connected simple cubic graphs on 22 vertices from Section 23.1. Each saved set has size at least 12. Every required edge-deletion witness was found and independently checked. There are 33 edges in each parent, hence 241,541,751 indexed parent-edge records in total. Only edges inside the saved good set require an additional mask; for every other edge that same good set already certifies a sufficiently large root-out state.

The producer `exploratory/edge_half_certificates.cpp` writes the parent graph6 string, its original mask and one deletion mask for each selected edge, in graph6 edge order. A separate checker, `exploratory/check_edge_half.cpp`, imports no search code. It independently decodes each graph, checks simplicity through the encoding, degree three, connectivity and the induced degree of every selected vertex. For each deletion mask it removes exactly the specified edge before checking degree two and the required cardinality. The Python driver `verify_round32.py` also compares every parent graph and original mask, row for row, against the retained order-22 census. All 64 shards pass. Enumeration completeness is inherited from the nauty census; the new witnesses are checked without trusting the optimization routine. In total, 95,430,602 new deletion masks are checked, and 146,111,149 parent-edge records reuse their original mask. An additional direct construction verifies good sets in 2,048 sampled mixed joins on 46 vertices.

Corollary 25.2 (Finite rooted family)**.** *Let Q_1, Q_2 be connected simple graphs on 23 vertices, each with one designated vertex of degree two and all other vertices of degree three. Suppose the two neighbours of each designated root are nonadjacent. Joining the roots of Q_1, Q_2 by an edge produces a cubic graph of order 46 with an induced 2-regular subgraph on at least 23 vertices.*

Proof. Suppressing either root produces a connected simple cubic graph on 22 vertices: its neighbours are distinct and nonadjacent, so suppression creates neither a loop nor a parallel edge. Each fragment is therefore an edge subdivision of a parent covered by the exhaustive check. Apply Proposition 25.1. □

This corollary is a computationally certified finite-family result. It includes symmetric and mixed joins, but not all cubic graphs of order 46. In particular, it excludes roots whose two neighbours are adjacent, and it gives no unrestricted assertion at higher orders. No counterexample to the original conjecture was obtained.

26 A normal form for rooted bridge obstructions

We now remove the triangle-at-the-root restriction from the finite result in Section 25. The structural reduction in this section applies at arbitrary orders; only its final census consequence is finite.

Let Q be a connected simple graph with one root r of degree two and every other vertex of degree three. Define

$$A = \max_{S \text{ good},\ r \in S} |S|, \quad B = \max_{S \text{ good},\ r \notin S} |S|, \quad C = c_{\mathrm{ind}}(Q - N_Q[r]),$$

using $A = -\infty$ when necessary. Put $d(Q) = c_{\mathrm{ind}}(D(Q)) = \max(2B, A + B)$. A *rooted obstruction* means $d(Q) < |V(Q)|$. This is a counterexample test for the cubic double, not a new version of the original conjecture.

26.1 Triangle expansion at the root

Replace the root r, whose neighbours are a, b, by a triangle $r'pqr'$, adding ap and bq, and take r' as the new root. Denote this operation by E; it adds two vertices.

Lemma 26.1 (Exact root expansion)**.** *The profile of $E(Q)$ is*

$$(A', B', C') = (C + 3,\ \max(B, A + 1),\ B).$$

Consequently the profile of $E^2(Q)$ is

$$(A'', B'', C'') = (B + 3,\ \max(B, A + 1, C + 4),\ \max(B, A + 1)).$$

Proof. A good set containing r' must contain p, q and avoid a, b. The remaining choices are precisely the good sets counted by C, so $A' = C + 3$. If r' is absent, either both p, q are absent or both are present: exactly one cannot have two selected neighbours. The first case contributes B. In the second, a, b must both be selected; replacing the selected path $apqb$ by arb gives a root-in good set of Q, smaller by one. Thus $B' = \max(B, A + 1)$. Finally, deleting the closed neighbourhood of r' leaves $Q - r$, whose optimum is B. Applying the same map again proves the second formula. □

Proposition 26.2 (Two expansions preserve the half bound)**.** *If $d(Q) \geq q = |V(Q)|$, then $d(E^2(Q)) \geq q + 4$.*

Proof. If $A > B$, then $d(Q) = A + B$, while $A'' + B'' \geq (B + 3) + (A + 1) = d(Q) + 4$. If $A \leq B$, including $A = -\infty$, then $d(Q) = 2B$. The order q is odd, so $2B \geq q$ implies $2B \geq q + 1$. Since $A'' + B'' \geq (B + 3) + B$, the new double has optimum at least $q + 4$ in this case too. Its required half bound is $q + 4$. □

This does not assert that a single root expansion preserves the bound. For a non-obstructing Q of order $2h + 1$, the exact failure criterion is

$$E(Q) \text{ is a rooted obstruction} \quad \Longleftrightarrow \quad B = h + 1, \qquad A \leq h, \qquad C \leq h - 2.$$

Indeed, failure forces $B' = \max(B, A + 1) \leq h + 1$, hence $A \leq h$. If $B \leq h$ too, the original double would already fail, so $B = h+1$. Now $A' + B' = C + h + 4 < 2h + 3$ is exactly $C \leq h - 2$. Conversely these three conditions make both $2B'$ and $A' + B'$ smaller than $2h + 3$. They describe a potential boundary obstruction; no graph realizing a counterexample profile has been found.

26.2 The diamond termination case

Given a rooted graph R with root b, add vertices r, u, v, a and edges

$$ru, rv, uv, ua, va, ab,$$

and designate r as the root. Write $L(R)$ for the resulting rooted graph. Its joining edge ab is a bridge, and its order is $|V(R)| + 4$.

Lemma 26.3. *If R has profile (A, B, C) and $M = \max(A, B)$, then $L(R)$ has profile*

$$(M + 3,\ \max(B + 3, M),\ M).$$

Moreover, $d(L(R)) \geq d(R) + 6$, and

$$d(E(L(R))) = 2M + 8.$$

Thus if R is not a rooted obstruction, neither $L(R)$ nor $E(L(R))$ is one.

Proof. No selected cycle can use the bridge ab. When r is selected, the triangle ruv is selected and a is absent, so any maximum good set in R is available. This gives $M + 3$. When r is absent, either no diamond vertex is selected, contributing M, or the triangle auv is selected, requiring b to be absent and contributing $B + 3$. Removing $N[r]$ leaves a with degree one; it cannot be selected, leaving optimum M in R.

Since $d(R) = M + B$, selecting a root-in optimum in one copy of $L(R)$ and a root-out optimum in the other gives $d(L(R)) \geq (M + 3) + (B + 3) = d(R) + 6$. The expansion map gives root-in optimum $M + 3$ and root-out optimum $M + 4$ for $E(L(R))$, proving its displayed double optimum. Finally $2M \geq d(R)$; the orders of $L(R)$ and $E(L(R))$ exceed that of R by four and six, respectively. These inequalities prove the last assertion. □

Theorem 26.4 (Rooted normal form)**.** *If a rooted obstruction exists, one of minimum order is obtained from a connected simple cubic graph by one of the following operations:*

1. *Subdivide one edge once and use the new vertex as root.*
2. *Replace an edge ab by ap, pq, qb, pr, qr, with fresh vertices p, q, r, and use r as root.*

Consequently, if any connected cubic counterexample with a bridge exists, a counterexample exists among symmetric doubles of these two kinds of rooted graphs.

Proof. Choose a rooted obstruction Q of minimum order. If its root's neighbours are nonadjacent, suppress the root. The result is a connected simple cubic graph, giving the first form.

Otherwise its root r belongs to a triangle ruv. Let a, b be the third neighbours of u, v. If $a = b$, this is precisely $L(R)$ for the smaller rooted component R on the other side of the bridge incident with a. Minimality and Lemma 26.3 give a contradiction.

If $a \neq b$, contract the triangle to a single degree-two root, obtaining a smaller connected simple rooted graph Q' with $Q = E(Q')$. If the neighbours of the root of Q' are nonadjacent, suppress that root to obtain the second stated form. If they are adjacent, inspect their third neighbours. When those are distinct, $Q' = E(Q'')$ for a smaller rooted graph Q''. Minimality makes Q'' non-obstructing, and Proposition 26.2 contradicts $Q = E^2(Q'')$. When those third neighbours coincide, $Q' = L(R)$ for a smaller rooted R; the expanded-diamond conclusion of Lemma 26.3 gives the contradiction. These cases exhaust the root triangle. The final assertion follows from bridge symmetrization and then minimum-order choice among rooted obstructions. It is an existence reduction, not a classification of all cubic graphs with bridges. □

26.3 Certification through rooted order twenty-three

We checked both normal forms at every edge of all 556,471 connected simple cubic graphs of even order four through twenty, giving 33,079,650 indexed rooted tests. For the first form at parent order twenty-two, the complete certificates from Section 25 supply the remaining cases. For a fragment of odd order q, each new record retains either a root-out good set of size at least $(q + 1)/2$, or a root-out and a root-in good set whose sizes sum to at least q. In either case the double satisfies its half bound.

The producer `root_normal_forms.cpp` and independent checker `check_root_normal_forms.cpp` reside in `exploratory/`. The checker reconstructs each fragment and directly verifies its degrees and selected-neighbour counts. The driver `verify_round33.py` checks generator and checker counts and the inherited certificate hashes. Census completeness relies on nauty. Separately, exhaustive vertex-subset enumeration on 84 small inputs verifies all four profile formulas, including the exact single-expansion failure criterion.

Corollary 26.5 (Finite bridge exclusion)**.** *Every connected rooted graph of the specified degree pattern on at most 23 vertices satisfies $d(Q) \geq |V(Q)|$. Any bridge join of two such rooted graphs satisfies the original half bound. In particular:*

1. *every connected cubic counterexample with a bridge has at least 30 vertices;*
2. *a cubic graph on 46 vertices with a bridge splitting it into two 23-vertex components satisfies the half bound, with no restriction on triangles at the bridge endpoints.*

Proof. If a rooted obstruction of order at most 23 existed, choose one of minimum order. The normal-form theorem would give a parent of order at most 22 in the first case or at most 20 in the second. All these cases have checked positive certificates, a contradiction. Mixed joins follow from bridge symmetrization. Every rooted fragment has odd order at least five: orders one and three cannot have the required degree-three vertices in a simple graph. Thus deleting a bridge from a connected cubic graph of order at most 28 leaves two rooted components each of order at most 23. This proves the first consequence; applying the same join statement to two 23-vertex components proves the second. □

27 Restricting the distinguished edge and its parent profile

The two forms in Theorem 26.4 can be restricted further. The distinguished edge must lie on a cycle; this does not require that the rest of the parent graph be bridgeless.

Proposition 27.1. *In a minimum-order rooted obstruction in either normal form, the replaced edge of the cubic parent is not a bridge.*

Proof. Suppose $e = ab$ is a bridge of the parent G. Its deletion gives rooted components R_1, R_2 of odd orders q_1, q_2. Write A_i, B_i for their root-in and root-out optima and $M_i = \max(A_i, B_i)$. These components are smaller than the putative rooted obstruction, so minimality gives

$$d(R_i) = M_i + B_i \geq q_i.$$

For the edge-subdivision form, the new root lies on no cycle. Its root-out optimum is $M_1 + M_2$. Since $2M_i \geq d(R_i) \geq q_i$ and q_i is odd, $2M_i \geq q_i + 1$. The double therefore has optimum at least $q_1 + q_2 + 2$, exceeding the required $q_1 + q_2 + 1$.

For the rooted-triangle form, both joining edges from the new triangle to R_1, R_2 are bridges. A root-in good set selects the triangle and must avoid both attachment roots, giving $A = 3 + B_1 + B_2$. A root-out good set selects no vertex of that triangle, giving $B = M_1 + M_2$. Thus

$$A + B = 3 + d(R_1) + d(R_2) \geq 3 + q_1 + q_2,$$

exactly the required half bound for its double. Both cases contradict rooted obstruction. □

27.1 Exact tests on the cubic parent

Let G have order $2h$, let $e = uv$, and define

$$U = c_{\mathrm{ind}}(G - e), \qquad T = \max_{S \text{ good in } G,\ u,v \in S} |S|, \qquad W = c_{\mathrm{ind}}(G - \{u, v\}).$$

As usual, an impossible constrained maximum is $-\infty$. The subdivision form Q_0 has profile $(T + 1, U, W)$. The triangle form $Q_1 = E(Q_0)$ consequently has profile

$$(A_1, B_1, C_1) = (W + 3,\ \max(U, T + 2),\ U).$$

These are direct instances of the root expansion formula.

Corollary 27.2. *If a minimum-order rooted obstruction has the triangle form, its parent edge satisfies*

$$U = h + 1, \qquad T \leq h - 1, \qquad W \leq h - 2.$$

Conversely, a cubic parent and edge satisfying these three conditions would give a counterexample to the original conjecture by doubling Q_1. Its order would be $4h + 6$.

Proof. Minimality makes Q_0, which has order $2h + 1$, non-obstructing. Apply the exact single-expansion failure criterion of Section 26.1 to $(A, B, C) = (T + 1, U, W)$ to get the three conditions. Conversely, they give $B_1 = h + 1$ and $A_1 \leq h + 1$, so the double has optimum at most $2h + 2$, below its required $2h + 3$. □

Every good set of the parent is either counted by T or survives the deletion of e. Consequently the displayed conditions imply $c_{\mathrm{ind}}(G) \leq h + 1$. If the parent satisfies the original half bound, its optimum must therefore be h or $h + 1$. Every good set of size at least h must contain exactly one endpoint of e: containing both contradicts the bound on T, and avoiding both contradicts the bound on W. If the parent is bipartite, every good set has even size, including in $G - e$. The equality $U = h + 1$ therefore forces h odd, so a bipartite parent of this minimum triangle form has order 2 modulo 4. These necessary conditions do not establish the existence or nonexistence of such a parent.

27.2 Targeted search and independent checks

Exhaustive vertex-subset checks on all 369 parent-edge cases from connected simple cubic graphs through order ten verify both profile identities directly. The bridge case in that census also verifies the component formulas used in Proposition 27.1.

We then sought a rooted graph Q_0 of order 25 satisfying $A \leq 12$, $B \leq 13$, $C \leq 10$. Such a graph would yield an original counterexample on 54 vertices after root expansion and doubling. The search fixes the two root neighbours, forbids their adjacency, and requires $Q_0 - r$ connected so that the suppressed parent edge is not a bridge. It permits deletion of at most six edges of a fixed seed, with degree constraints forcing the same number of replacements. The seed comes from subdividing a non-bridge edge of a retained sharp 24-vertex cubic parent.

A counterexample-guided satisfiability search proposes graphs and exhaustively enumerates good sets of size at least eleven. Every violating set adds a constraint excluding that selected set from being good in subsequent candidates. The three thresholds detect all violations of the stated target bounds. Both the initial rooted double and its expanded double are also tested, to avoid missing an earlier counterexample. A separate cycle-space verifier checks the exact profiles of all retained connected candidates. The search is restricted to a labelled neighbourhood of one seed; a timeout or an unknown solver result does not exhaust this domain. The source is `exploratory/root_triangle_synthesis.py`, and the independent verifier is `verify_round34.py`.

The run reached its wall-time stopping condition after 407.4 seconds, with 12 retained candidates and 1,919 exclusion constraints, including 160 initial constraints. The wall-time condition is checked between iterations. Independent enumeration of all 8,192 cycle-space vectors for each candidate confirms its exact profile. The minimum doubled optima among these candidates are 30 before expansion and 32 after expansion, compared with required half bounds of 25 and 27. No candidate is a counterexample. The remaining search region was not exhausted, and no general sufficiency proof follows from the run.

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